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Kim et al.

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(54) **MEMORY DEVICE INCLUDING ROW DECODER**

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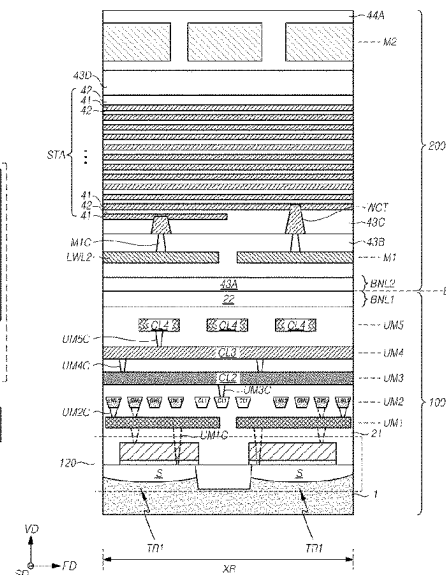
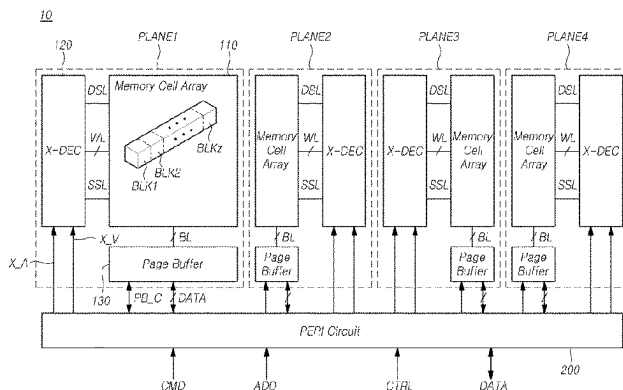
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Primary Examiner — Tha-O H Bui

(57) **ABSTRACT**

A memory device includes a first semiconductor structure including pass transistors defined in a row decoder region of a substrate, a first bonding layer including first bonding pads, and bottom wiring layers disposed between the substrate and the first bonding layer; a second semiconductor structure including a second bonding layer including second bonding pads bonded to the first bonding pads, a memory cell array, and a top wiring layer disposed between the second bonding layer and the memory cell array; and global lines disposed in the row decoder region, and configured to transfer operating voltages to the pass transistors, wherein the bottom wiring layers include bottom wiring layers of a first tier and bottom wiring layers of a second tier disposed over the bottom wiring layers of the first tier, and the global lines are disposed in at least one of the bottom wiring layers of the first tier.

19 Claims, 21 Drawing Sheets



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FIG. 1

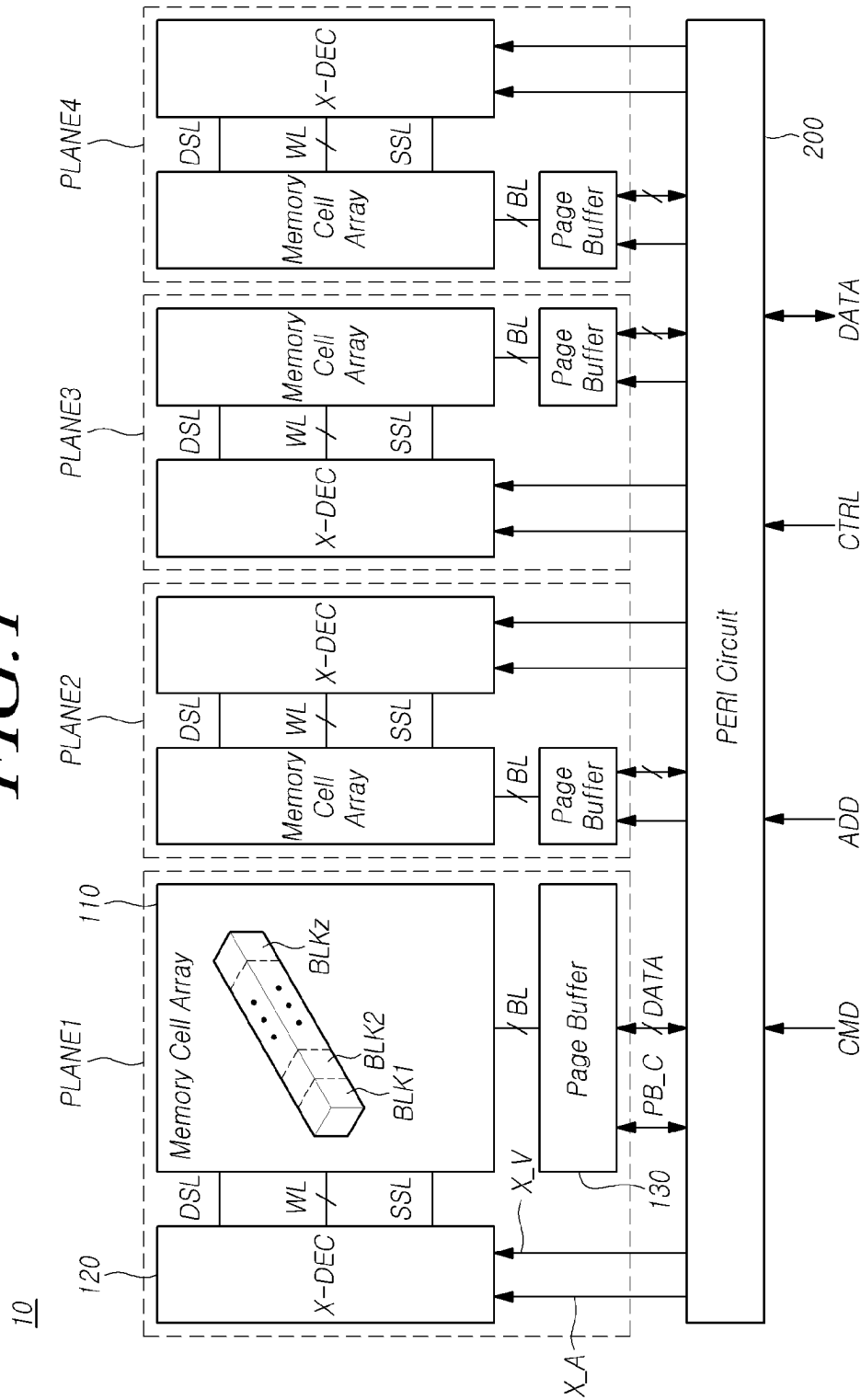


FIG. 2

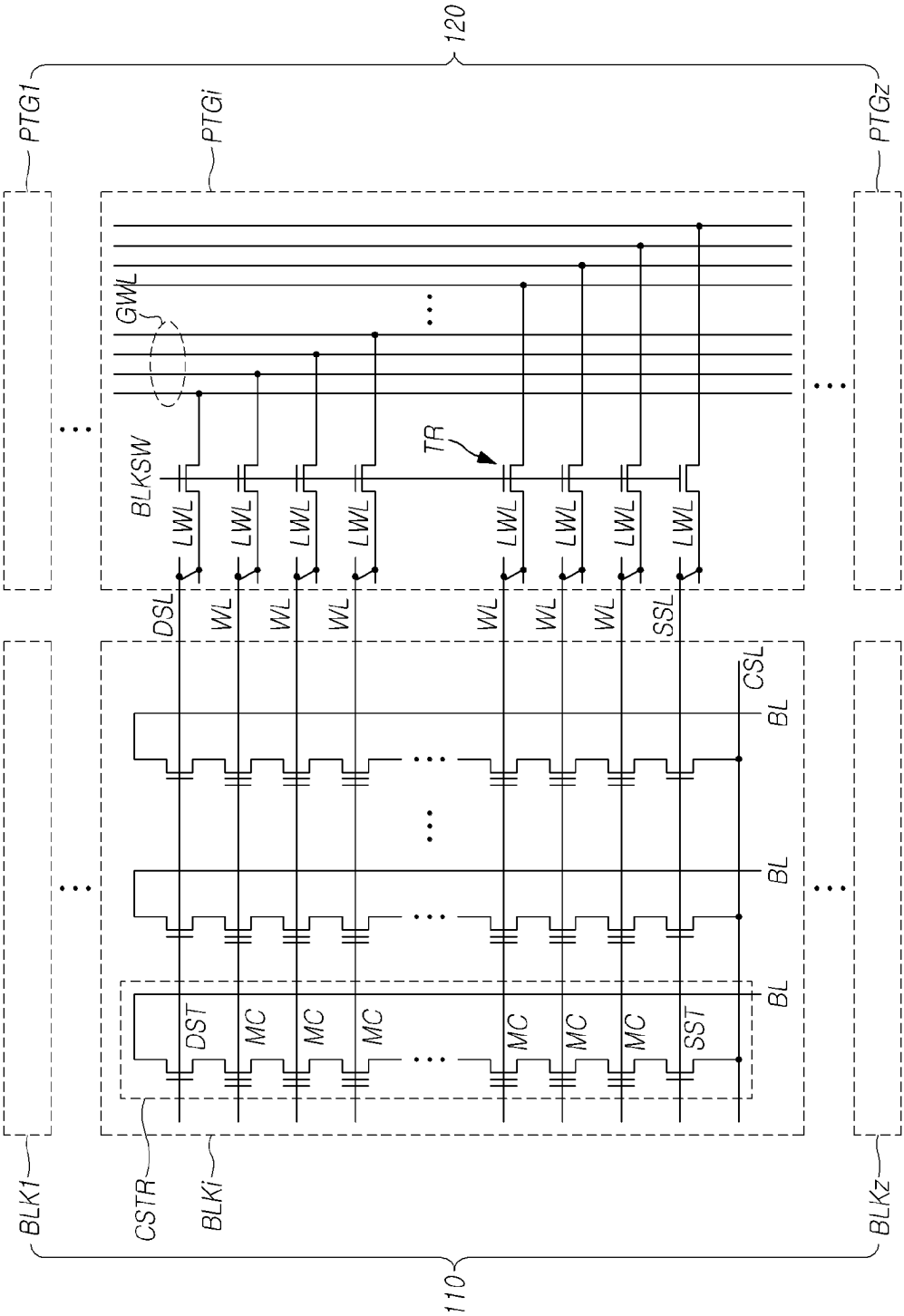


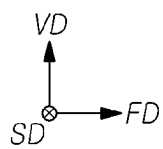
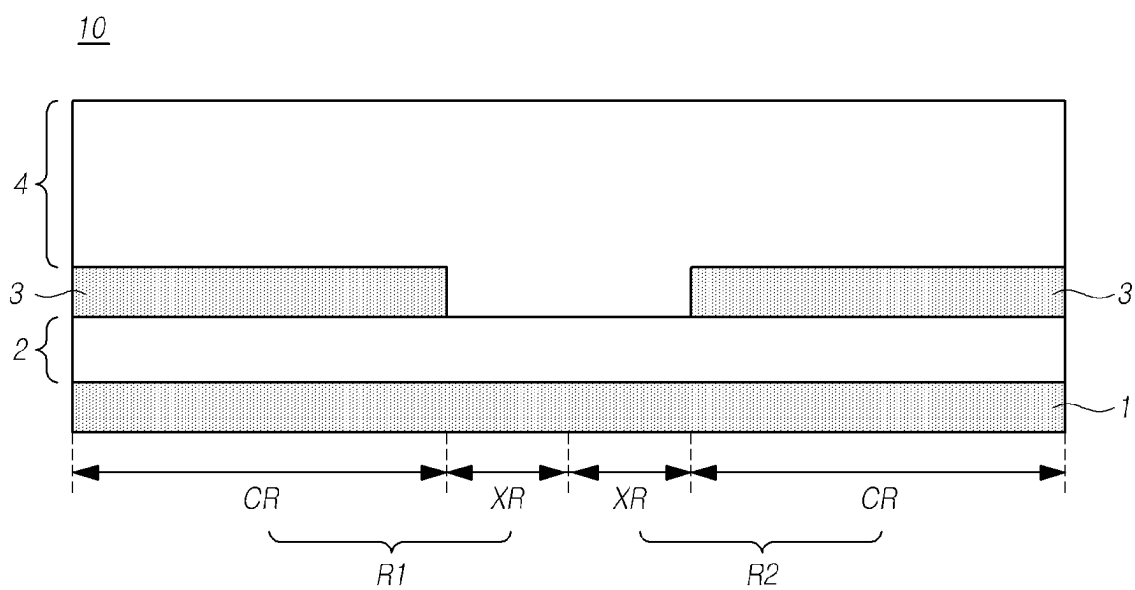
FIG. 3

FIG. 4

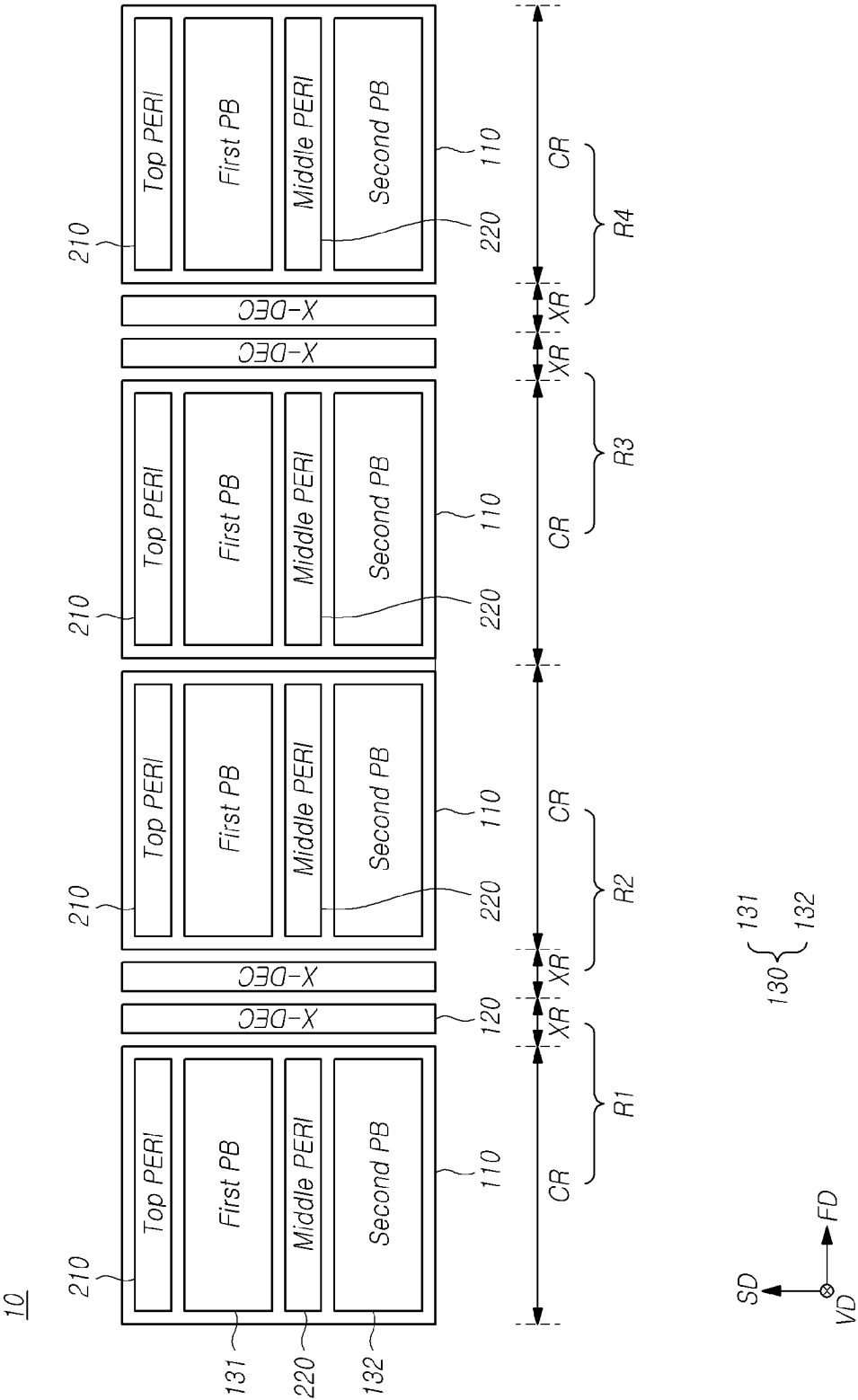


FIG. 6

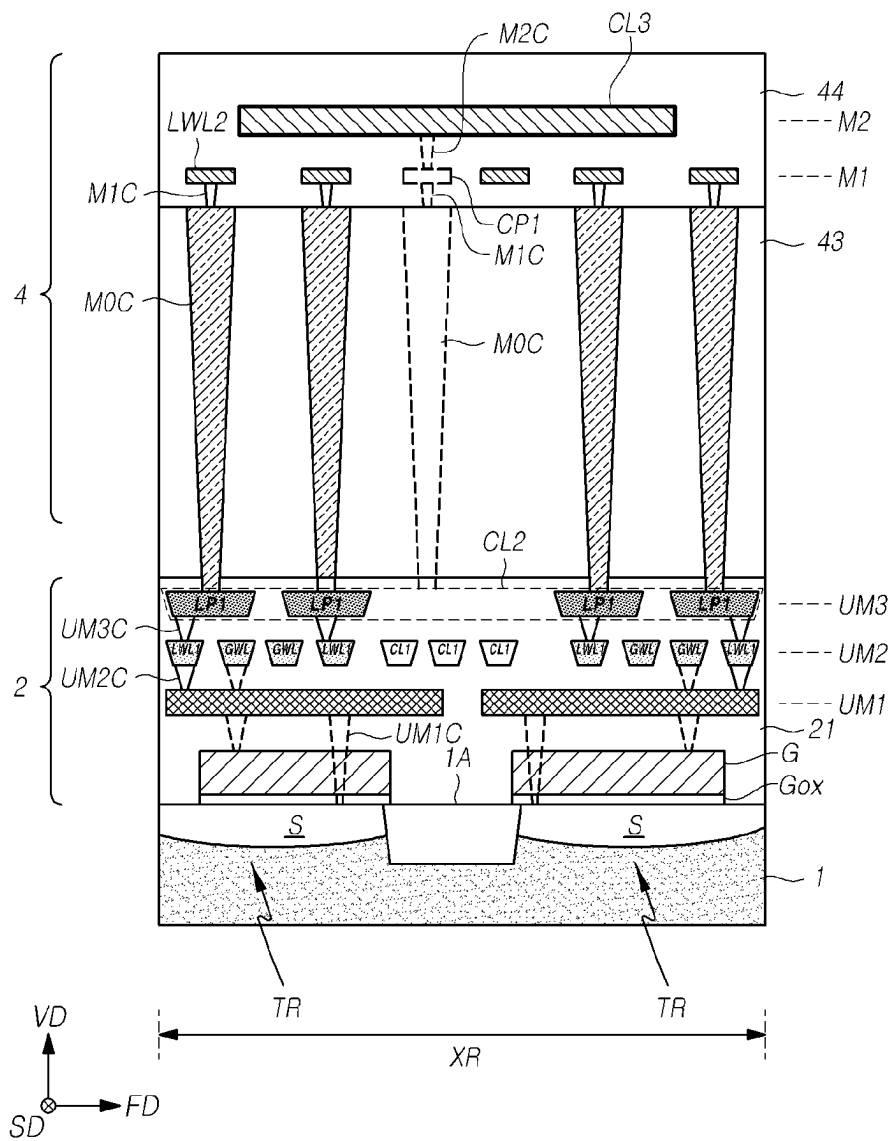


FIG. 7A

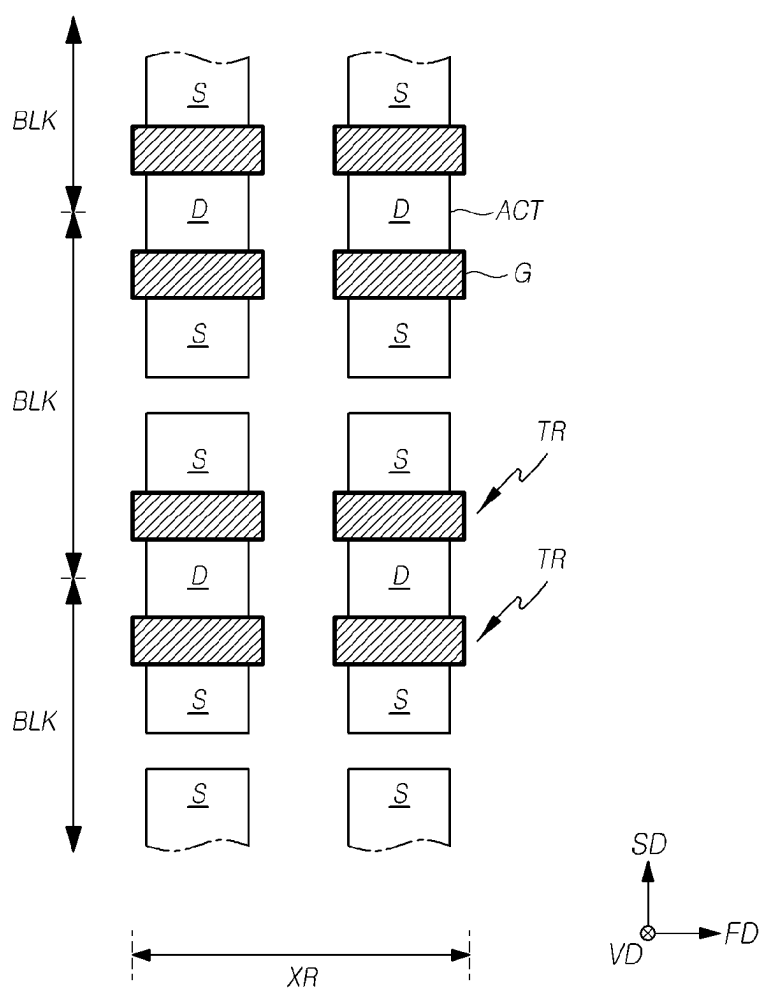


FIG. 7C

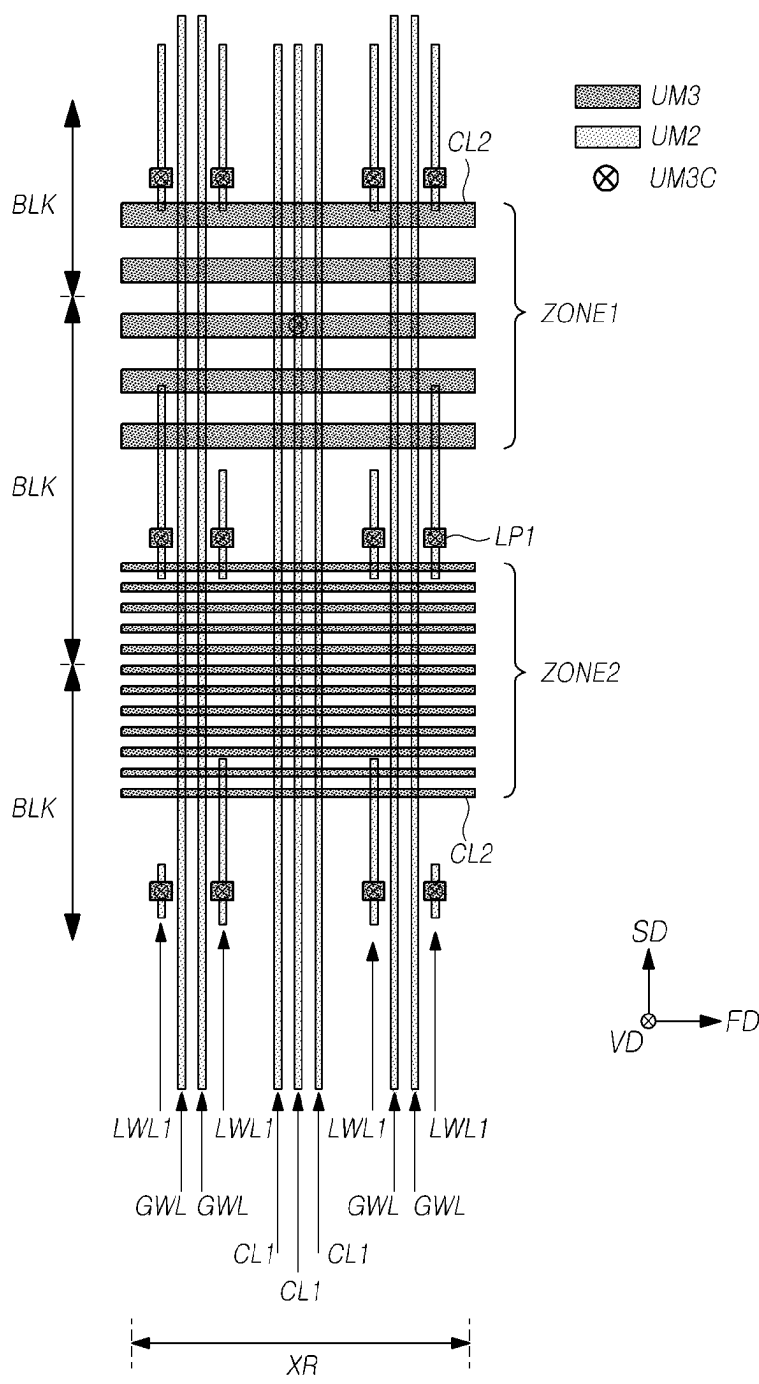


FIG. 7D

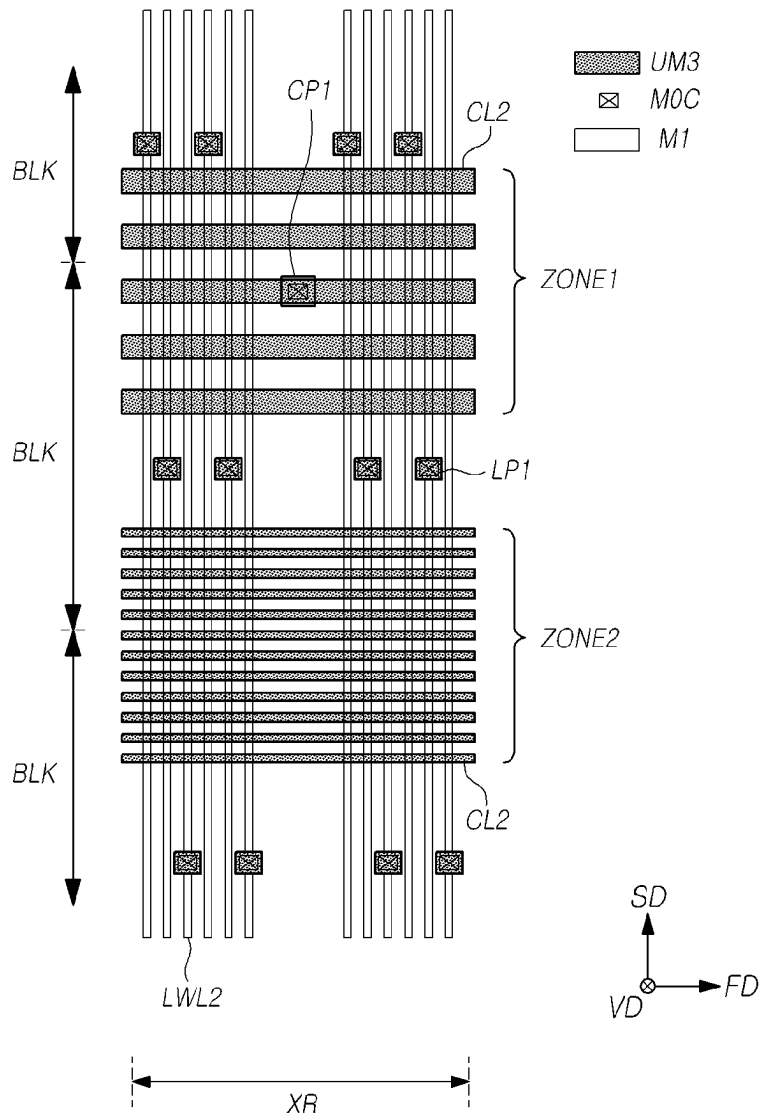


FIG. 7E

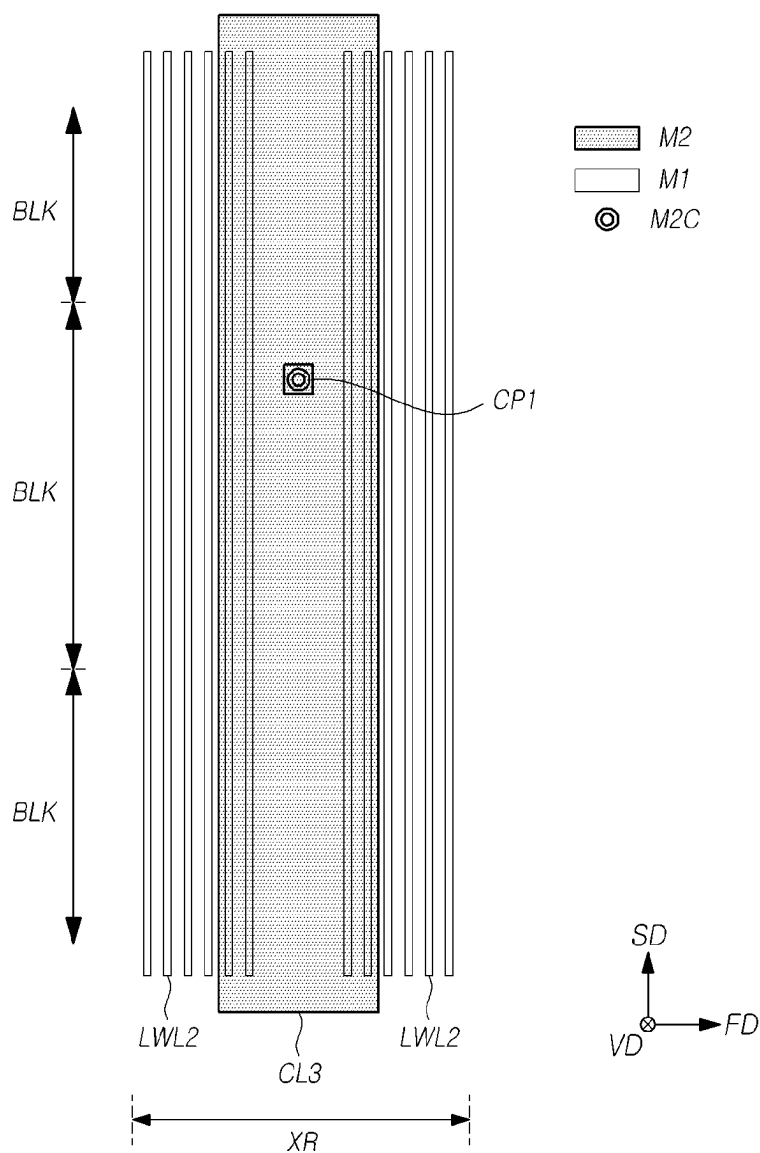


FIG. 9

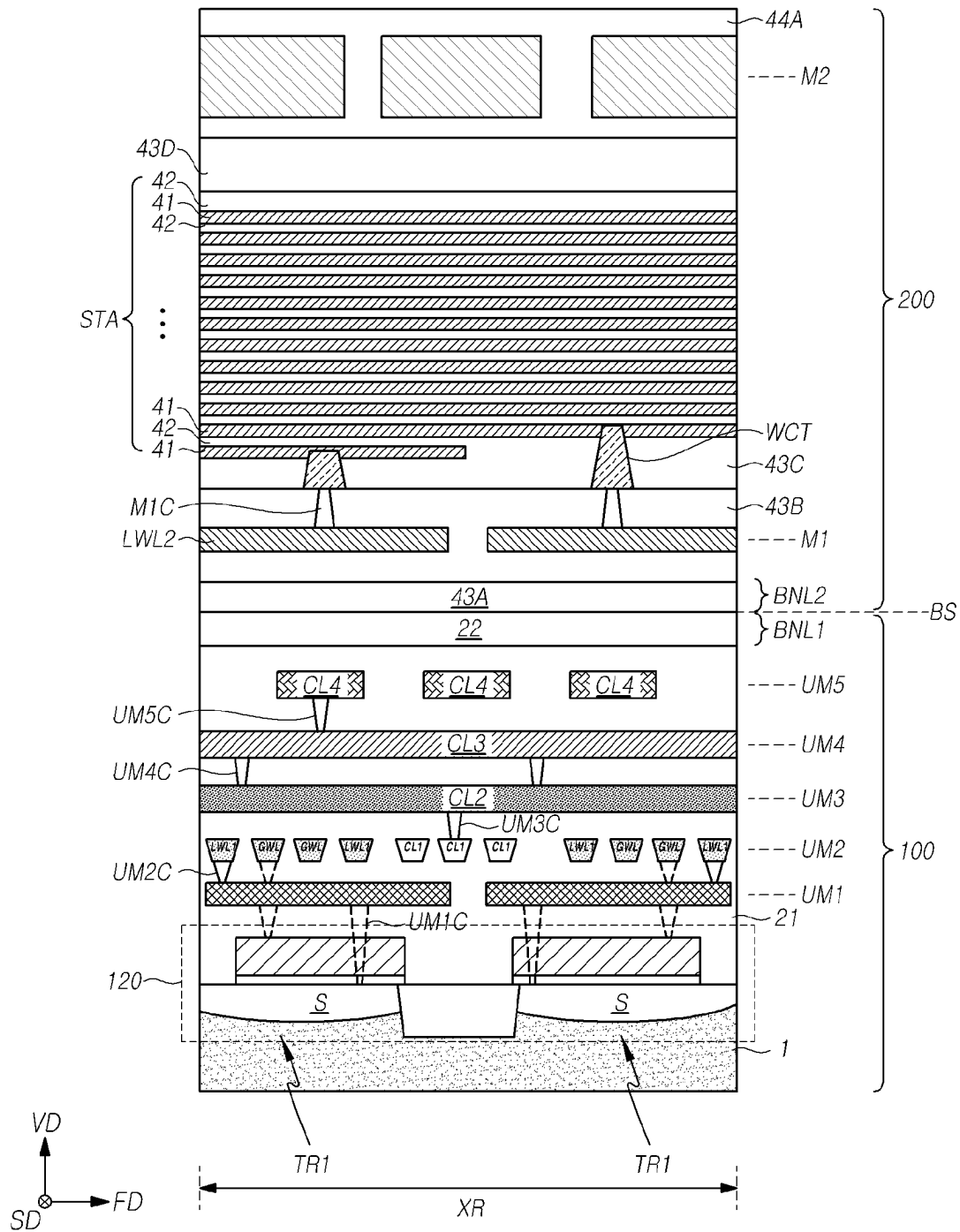


FIG. 10

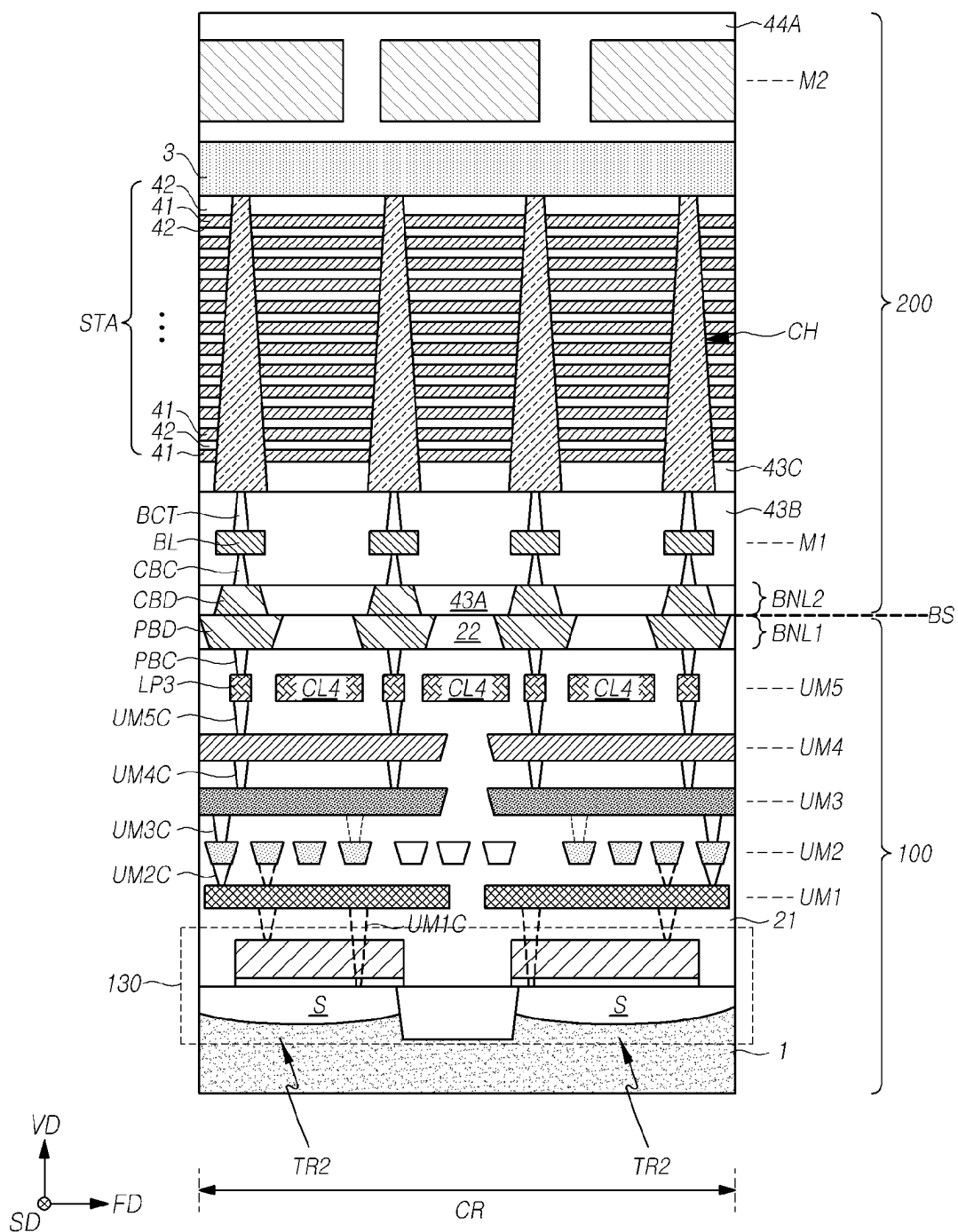


FIG. 11A

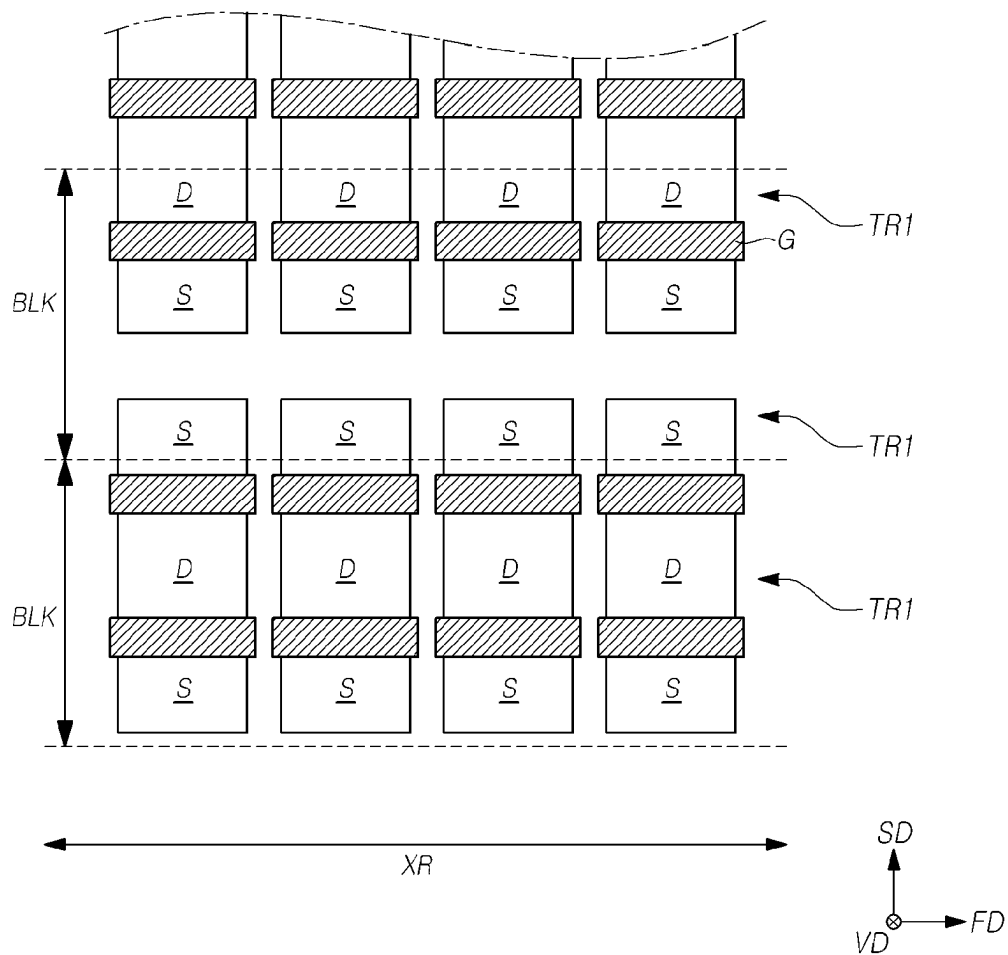


FIG. 11B

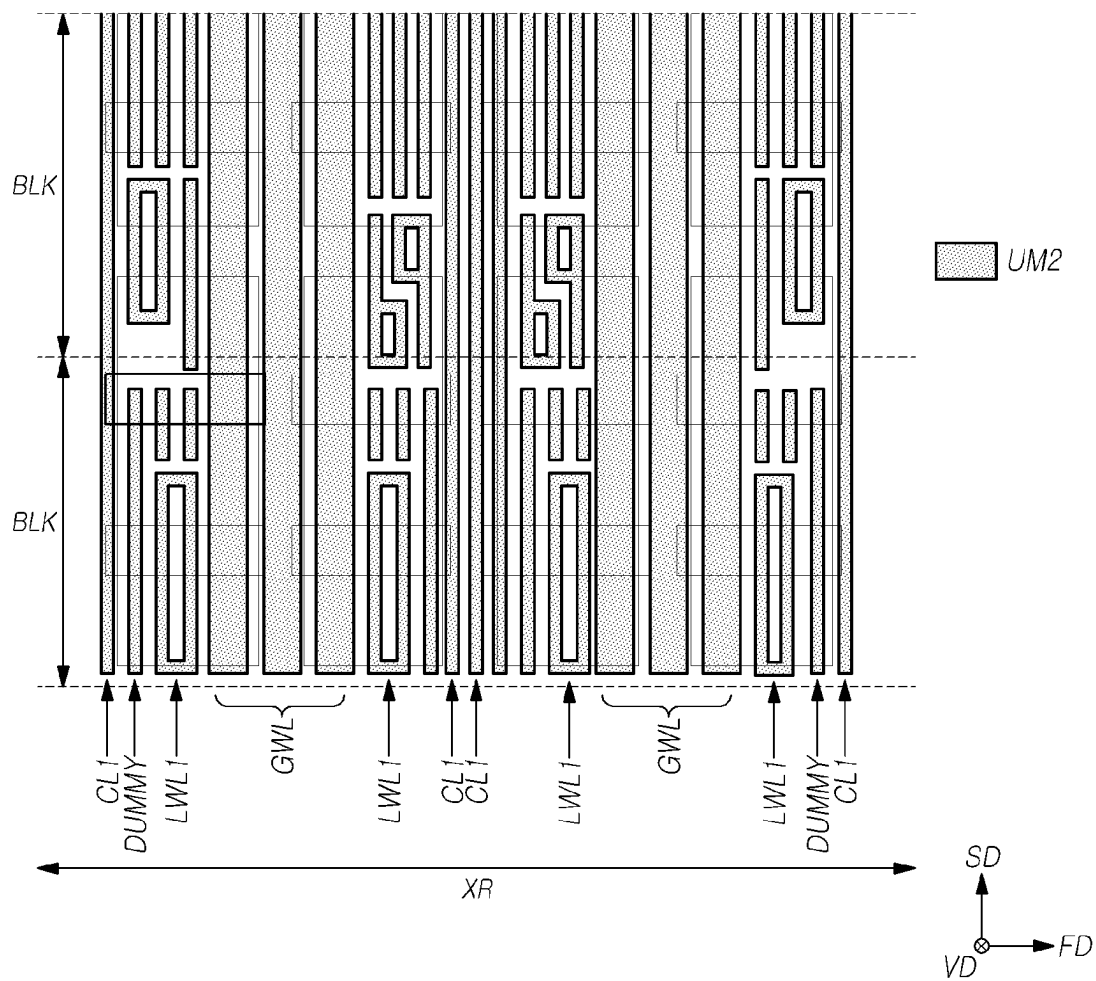


FIG. 11C

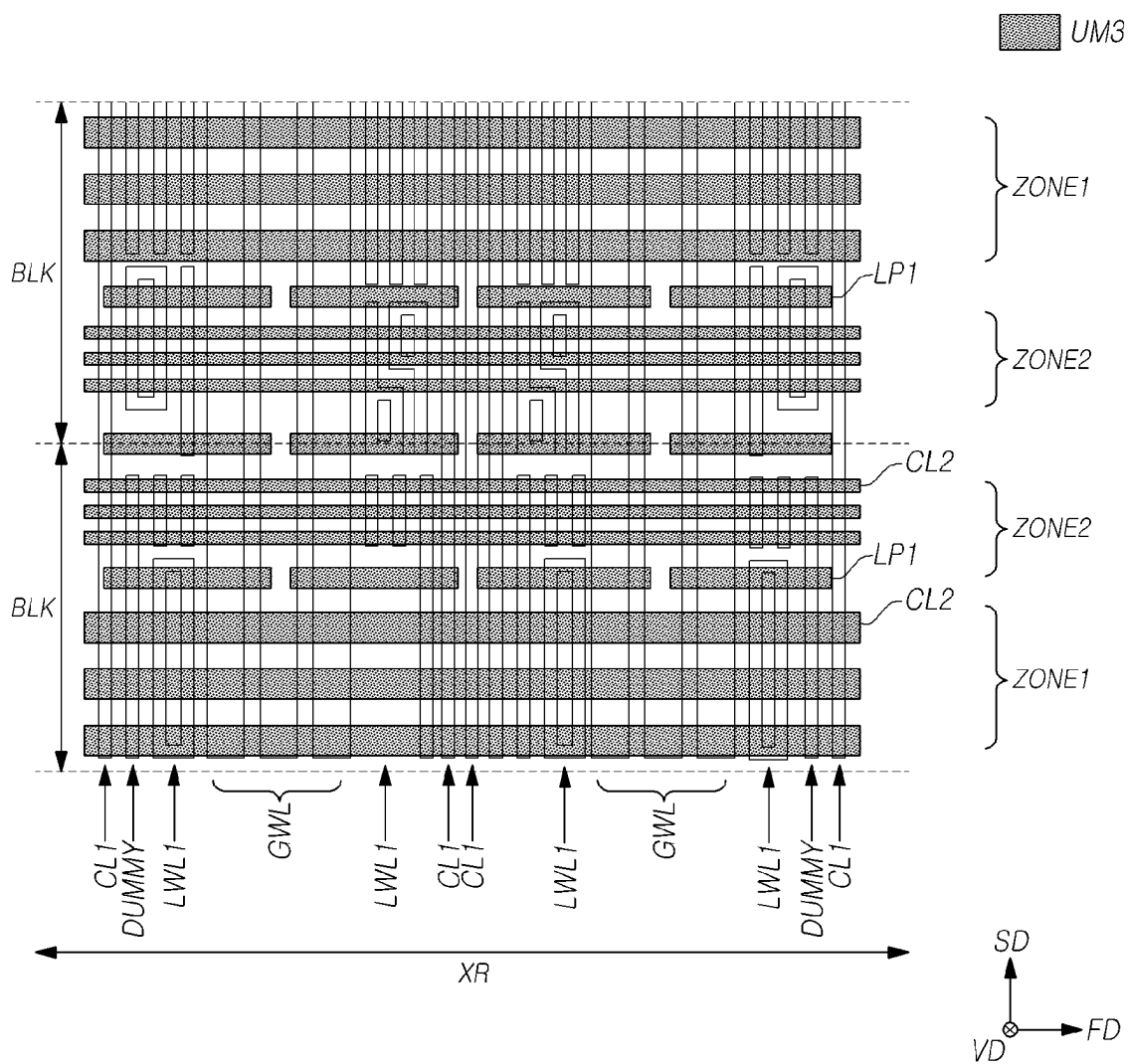


FIG. 11D

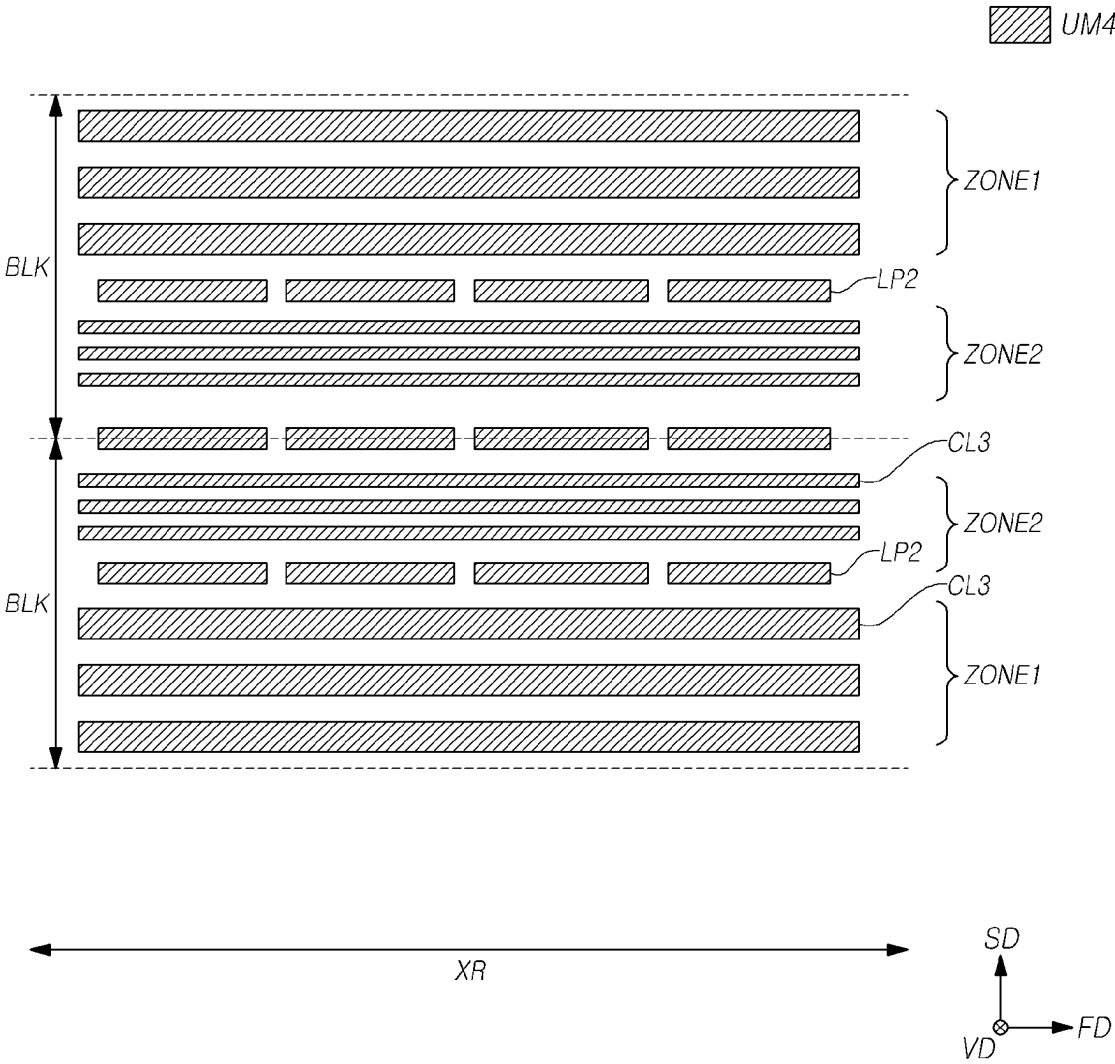


FIG. 11E

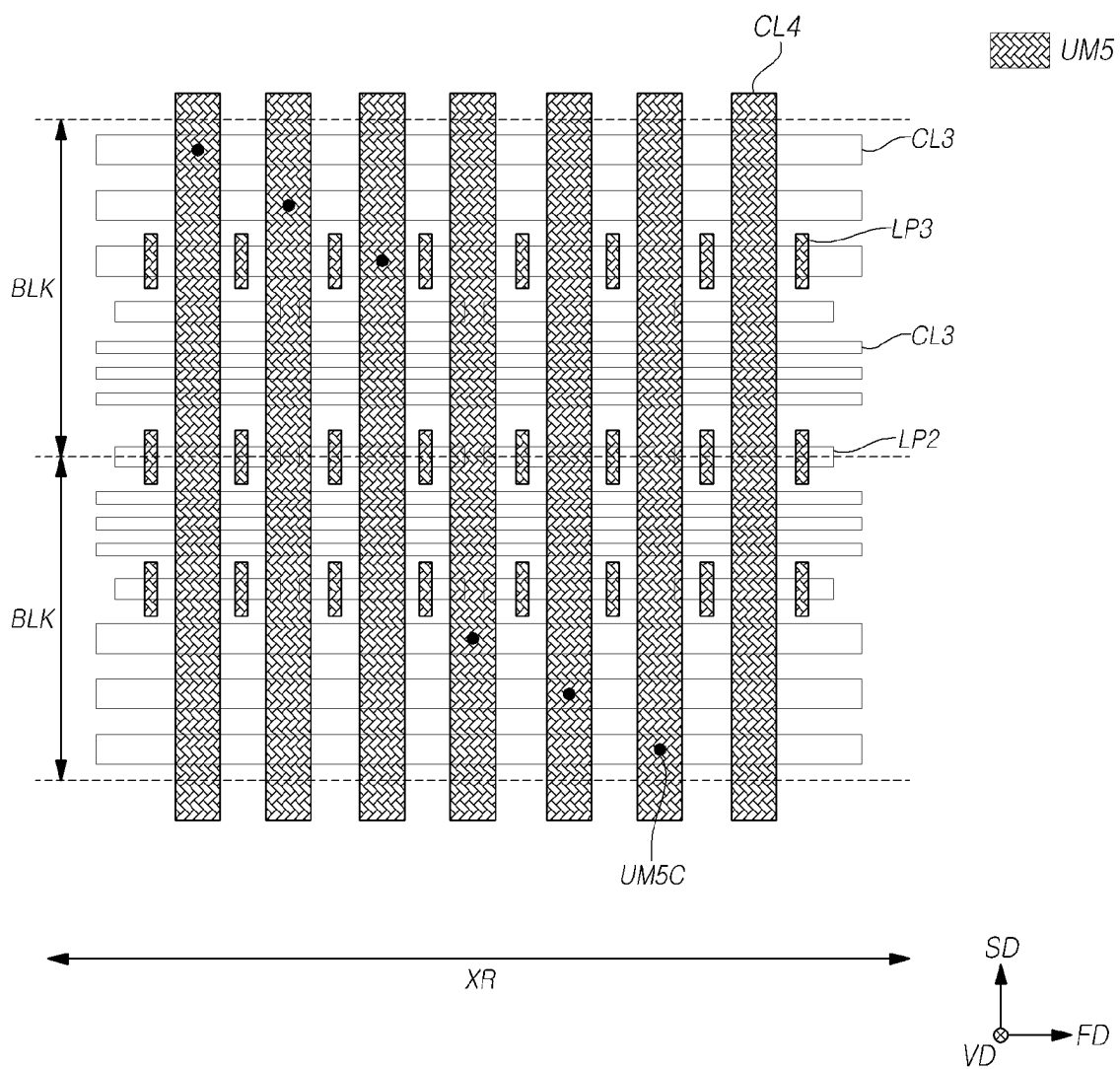


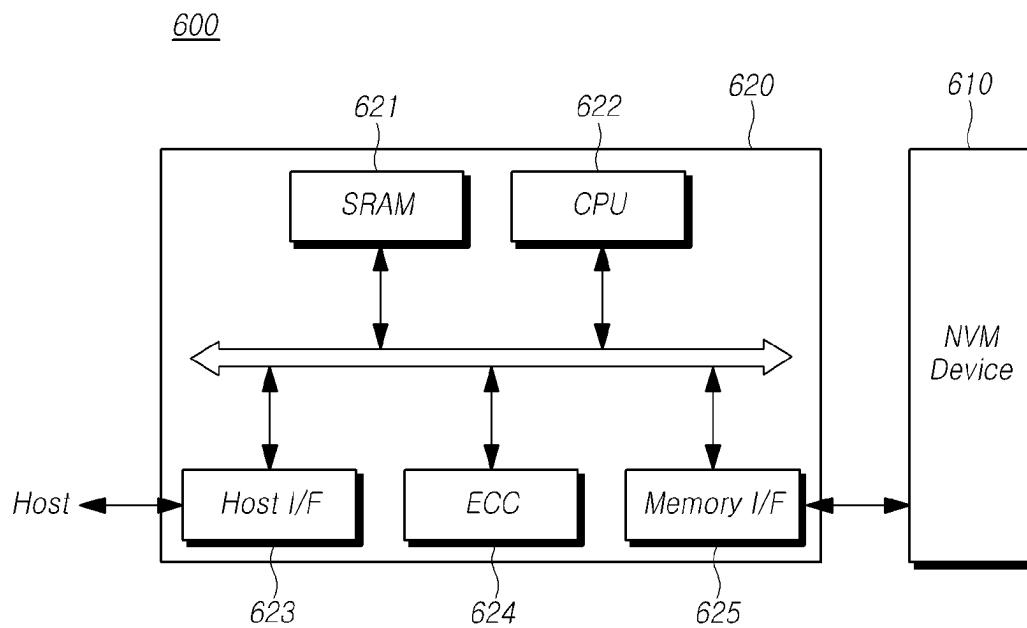
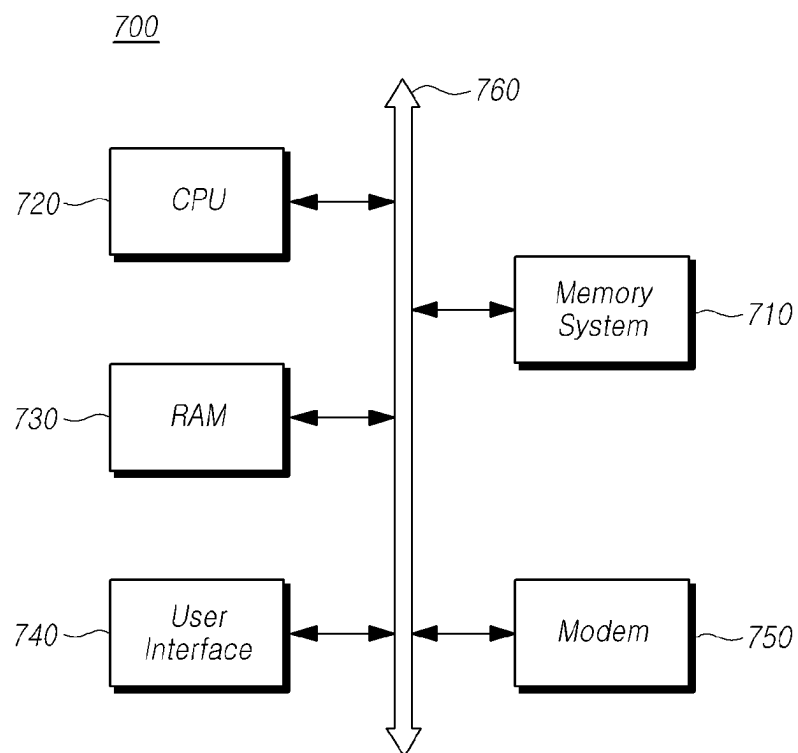
FIG. 12

FIG. 13

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MEMORY DEVICE INCLUDING ROW DECODER

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation-in-part of pending U.S. patent application Ser. No. 17/165,097, filed on Feb. 2, 2021, which claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2020-0114973 filed in the Korean Intellectual Property Office on Sep. 8, 2020, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

Various embodiments generally relate to a semiconductor technology, and more particularly, to a memory device that includes a row decoder.

2. Related Art

A volatile memory device has high write and read speeds, but may lose data stored therein if its power supply is interrupted. A nonvolatile memory device has relatively low write and read speeds, but may retain data stored therein even though its power supply is interrupted. Therefore, in order to store data that should be retained regardless of power supply, a nonvolatile memory device may be used. Examples of nonvolatile memory devices include a read only memory (ROM), a mask ROM (MROM), a programmable ROM (PROM), an erasable programmable ROM (EPROM), an electrically erasable and programmable ROM (EEPROM), a flash memory, a phase change random access memory (PRAM), a magnetic RAM (MRAM), a resistive RAM (RRAM), and a ferroelectric RAM (FRAM). Flash memories may be divided into NOR type and NAND type memories.

Among the examples of nonvolatile memory devices, a NAND flash memory device is widely used as a data storage device. The NAND flash memory device may transfer operating voltages to memory cells through a row decoder.

SUMMARY

Various embodiments are directed to measures capable of reducing the size and failure of a memory device.

In an embodiment, a memory device may include: a first semiconductor structure including a peripheral circuit which is defined in a cell region of a substrate, a plurality of pass transistors which are defined in a row decoder region of the substrate, a first bonding layer which includes a plurality of first bonding pads, and a plurality of bottom wiring layers which are disposed between the substrate and the first bonding layer; a second semiconductor structure disposed on the first semiconductor structure, and including a second bonding layer which includes a plurality of second bonding pads bonded to the plurality of first bonding pads, a memory cell array, and at least one top wiring layer which is disposed between the second bonding layer and the memory cell array; and a plurality of global lines disposed in the row decoder region, and configured to transfer operating voltages to the plurality of pass transistors, wherein the plurality of bottom wiring layers include bottom wiring layers of a first tier and bottom wiring layers of a second tier which are disposed over the bottom wiring layers of the first tier, and

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wherein the plurality of global lines are disposed in at least one of the bottom wiring layers of the first tier.

In an embodiment, a memory device may include: a first semiconductor structure including a substrate which includes first and second cell regions adjacent to each other in a first direction and a row decoder region between the first cell region and the second cell region, a peripheral circuit which is defined in the first and second cell regions of the substrate, a plurality of pass transistors which are defined in the row decoder region of the substrate, a first bonding layer which includes a plurality of first bonding pads, and a plurality of at least three bottom wiring layers which are disposed between the substrate and the first bonding layer; a second semiconductor structure disposed on the first semiconductor structure, and including a second bonding layer which includes a plurality of second bonding pads bonded to the plurality of first bonding pads, a memory cell array, and at least one top wiring layer which is disposed between the second bonding layer and the memory cell array; and a plurality of global lines disposed in the row decoder region, and configured to transfer operating voltages to the plurality of pass transistors, wherein the plurality of global lines are disposed in at least one of a first bottom wiring layer which is positioned at a lowermost part among the plurality of bottom wiring layers and a second bottom wiring layer immediately over the first bottom wiring layer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a representation of a memory device in accordance with an embodiment of the disclosure.

FIG. 2 is a circuit diagram illustrating a representation of a memory cell array and a row decoder illustrated in FIG. 1.

FIG. 3 is a cross-sectional view illustrating a representation of a schematic layout of a memory device in accordance with an embodiment of the disclosure.

FIG. 4 is a top view illustrating a representation of a layout of a memory device in accordance with an embodiment of the disclosure.

FIGS. 5 and 6 are cross-sectional views illustrating a representation of a memory device in accordance with an embodiment of the disclosure.

FIGS. 7A to 7E are top views illustrating representations of main components of a memory device in accordance with an embodiment of the disclosure.

FIGS. 8 and 9 are cross-sectional views illustrating a representation of a row decoder region of a memory device in accordance with an embodiment of the disclosure.

FIG. 10 is a cross-sectional view illustrating a representation of a cell region of the memory device in accordance with the embodiment of the disclosure.

FIGS. 11A to 11E are top views illustrating representations of main components of a memory device in accordance with an embodiment of the disclosure.

FIG. 12 is a block diagram schematically illustrating a representation of a memory system including a memory device in accordance with an embodiment of the disclosure.

FIG. 13 is a block diagram schematically illustrating a representation of a computing system including a memory device in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

Advantages and features of the disclosure and methods to achieve them will become apparent from the descriptions of exemplary embodiments herein below and described with

reference to the accompanying drawings. However, the present disclosure is not limited to the exemplary embodiments disclosed herein but may be implemented in various different ways. The exemplary embodiments of the present disclosure convey the scope of the disclosure to those skilled in the art.

Because the figures, dimensions, ratios, angles, numbers of elements given in the drawings that describe embodiments of the disclosure are merely illustrative, the present disclosure is not limited to the illustrated matters. Throughout the specification, like reference numerals refer to like components. In describing the disclosure, when it is determined that a detailed description of the related art may obscure the gist or clarity of the disclosure, the detailed description thereof will be omitted. It is to be understood that the terms “comprising,” “having,” “including” and so on, used in the description and claims, should not be interpreted as being restricted to the means listed thereafter unless specifically stated otherwise. Where an indefinite or definite article (e.g., “a,” “an” or “the”) is used when referring to a singular noun, the article may include a plural of that noun unless specifically stated otherwise.

In interpreting elements in embodiments of the disclosure, they should be interpreted as including error margins even in the absence of explicit statements.

Also, in describing the components of the disclosure, there may be used terms such as first, second, A, B, (a), and (b). These are solely for the purpose of differentiating one component from another component and do not limit the substances, order, sequence or number of the components. Also, components in embodiments of the disclosure are not limited by these terms. These terms are used to merely distinguish one component from another component. Accordingly, as used herein, a first component may be a second component within the technical spirit of the disclosure.

If a component is described as “connected,” “coupled” or “linked” to another component, it may mean that the component is not only directly “connected,” “coupled” or “linked” but also is indirectly “connected,” “coupled” or “linked” via a third component. In describing positional relationship, such as “an element A on an element B,” “an element A above an element B,” “an element A below an element B” and “an element A next to an element B,” one or more other elements may be disposed between the elements A and B unless the term “directly” or “immediately” is explicitly used.

Features of various exemplary embodiments of the disclosure may be coupled, combined or separated partially or totally. Technically various interactions and operations are possible. Various exemplary embodiments can be practiced individually or in combination.

Hereinafter, various examples of embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

FIG. 1 is a block diagram illustrating a representation of a memory device in accordance with an embodiment of the disclosure.

Referring to FIG. 1, a memory device 10 in accordance with an embodiment of the disclosure may include a plurality of planes PLANE1 to PLANE4 and a peripheral circuit (PERI Circuit) 200. The memory device 10 may have a four-plane structure. While the embodiment of FIG. 1 illustrates a memory device 10 with four planes, it is to be noted that the number of planes included in the memory device 10 is not limited thereto.

Each of the planes PLANE1 to PLANE4 may be substantially the same and may include a memory cell array 110, a row decoder (X-DEC) 120 and a page buffer circuit (Page Buffer) 130.

The memory cell array 110 may include a plurality of memory blocks BLK1 to BLKz (where z is a natural number of 2 or more). Each of the memory blocks BLK1 to BLKz may be coupled to the row decoder 120 through at least one drain select line DSL, a plurality of word lines WL and at least one source select line SSL. The memory blocks BLK1 to BLKz may be coupled to the page buffer circuit 130 through a plurality of bit lines BL. The drain select line DSL, the word lines WL and the source select line SSL may be provided for each of the memory blocks BLK1 to BLKz, and the bit lines BL may be provided in common for the memory blocks BLK1 to BLKz.

Each of the memory blocks BLK1 to BLKz may include a plurality of cell strings. Although not illustrated, each of the cell strings may include, coupled in series, at least one drain select transistor, a plurality of memory cells, and at least one source select transistor.

The row decoder 120 may select any one from among the memory blocks BLK1 to BLKz included in the memory cell array 110, in response to a row address X_A provided from the peripheral circuit 200. The row decoder 120 may transfer an operating voltage X_V, provided from the peripheral circuit 200, to the drain select line DSL, the word lines WL and the source select line SSL, which are coupled to a selected memory block.

The page buffer circuit 130 may include a plurality of page buffers (not illustrated) that are coupled to the plurality of bit lines BL, respectively. The page buffer may receive a page buffer control signal PB_C from the peripheral circuit 200, and may transmit and receive a data signal DATA to and from the peripheral circuit 200. The page buffer may control the bit line BL in response to the page buffer control signal PB_C. For example, the page buffer may detect data, stored in a memory cell of the memory cell array 110, by sensing a signal of the bit line BL in response to the page buffer control signal PB_C, and may transmit the data signal DATA to the peripheral circuit 200 depending on the detected data. The page buffer may apply a signal to the bit line BL, based on the data signal DATA received from the peripheral circuit 200, in response to the page buffer control signal PB_C, and thereby, may write data to a memory cell of the memory cell array 110. The page buffer may write data to or read data from a memory cell that is coupled to an activated word line WL.

The peripheral circuit 200 may receive a command signal CMD, an address signal ADD and a control signal CTRL from outside the memory device 10, and may transmit and receive data DATA to and from a device outside the memory device 10, for example, a memory controller. The peripheral circuit 200 may output signals for writing data to the memory cell array 110 or reading data from the memory cell array 110, for example, the row address X_A, the page buffer control signal PB_C and so forth, based on the command signal CMD, the address signal ADD and the control signal CTRL. The peripheral circuit 200 may generate various voltages, including the operating voltage X_V, which are required in the memory device 10.

Although, in the present specification, a case where the memory device 10 is a flash memory is described as an example, in other embodiments, the type of a memory is not limited thereto, and the disclosure may also be applied to other memories in addition to flash memories. For example, a memory may be any one from among a dynamic random

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access memory (DRAM), a phase change RAM (PRAM), a magnetic RAM (MRAM), a resistive RAM (RRAM) and an ferroelectric RAM (FRAM).

FIG. 2 is a circuit diagram illustrating a representation of a memory cell array and a row decoder illustrated in FIG. 1.

Referring to FIG. 2, a memory cell array **110** included in each of a plurality of planes **PLANET** to **PLANE4** may include a plurality of memory blocks **BLK1** to **BLKz**. Each of the memory blocks **BLK1** to **BLKz** may correspond to an erase unit when a memory device performs an erase operation.

Each of the memory blocks **BLK1** to **BLKz** may include a plurality of cell strings **CSTR**, which are coupled between a plurality of bit lines **BL** and a common source line **CSL**. As illustrated in FIG. 2, in a memory block **BLKi**, each of the cell strings **CSTR** may include a drain select transistor **DST**, which is coupled to a bit line **BL**, a source select transistor **SST**, which is coupled to a common source line **CSL**, and a plurality of memory cells **MC**, which are coupled in series between the drain select transistor **DST** and the source select transistor **SST**.

The plurality of cell strings **CSTR** included in each of the memory blocks **BLK1** to **BLKz** may be coupled in common to the common source line **CSL**. Each of the cell strings **CSTR** may be coupled to a corresponding bit line **BL**.

The gates of drain select transistors **DST** may be coupled to a drain select line **DSL**. The gates of each of memory cells **MC** may be coupled to a corresponding word line **WL**. The gates of source select transistors **SST** may be coupled to a source select line **SSL**. The drain select line **DSL**, the word lines **WL** and the source select line **SSL** may be coupled to local lines **LWL**, respectively.

A row decoder **120** may include a plurality of pass transistor groups **PTG1** to **PTGz** corresponding to the memory blocks **BLK1** to **BLKz**, respectively. Each of the plurality of pass transistor groups **PTG1** to **PTGz** may include a plurality of pass transistors **TR**, which are coupled between global lines **GWL** and the local lines **LWL**. The pass transistors **TR** may transfer operating voltages, applied to the global lines **GWL**, to a corresponding memory block through the local lines **LWL** in response to a block select signal **BLKSW**. The local lines **LWL** may be provided for each of the pass transistor groups **PTG1** to **PTGz**, and the global lines **GWL** may be provided in common for the pass transistor groups **PTG1** to **PTGz**. Thus, the pass transistor groups **PTG1** to **PTGz** may share the global lines **GWL**.

Hereinbelow, in the accompanying drawings, two directions that are parallel to the top surface of a substrate and intersect with each other are defined as a first direction **FD** and a second direction **SD**, respectively, and a direction that vertically protrudes from the top surface of the substrate is defined as a vertical direction **VD**. For example, the first direction **FD** may correspond to the extending direction of word lines, and the second direction **SD** may correspond to the extending direction of bit lines. The first direction **FD** and the second direction **SD** may substantially perpendicularly intersect with each other. In the drawings, a direction indicated by an arrow and a direction opposite thereto represent the same direction.

FIG. 3 is a cross-sectional view illustrating a representation of a schematic layout of a memory device in accordance with an embodiment of the disclosure.

Referring to FIG. 3, a memory device **10** in accordance with an embodiment of the disclosure may have a PUC (peri under cell) structure. The memory device **10** may include a substrate **1**, a logic structure **2** that is disposed on the substrate **1**, source plates **3** that are disposed on the logic

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structure **2**, and a memory structure **4** that is disposed on the logic structure **2** and the source plates **3**.

The substrate **1** of the memory device **10** may include a first plane region **R1** and a second plane region **R2**, which are disposed in the first direction **FD**. Each of the first and second plane regions **R1** and **R2** may include a cell region **CR** and a row decoder region **XR** that is adjacent to the respective cell region **CR** in the first direction **FD**.

As will be described later, the logic structure **2** may include row decoders, page buffer circuits, and a peripheral circuit. Further, the logic structure **2** may include a plurality of bottom wiring layers, which are positioned at different height levels.

The source plates **3** may be disposed in the cell region **CR** of the first plane region **R1** and the cell region **CR** of the second plane region **R2**, respectively.

As will be described later, the memory structure **4** may include memory cell arrays that are disposed on the source plates **3**, respectively, and a plurality of top wiring layers, which are positioned at different height levels. FIG. 3 illustrates a structure that includes two planes. The memory cell array that is disposed in the first plane region **R1** may configure a first plane, and the memory cell array that is disposed in the second plane region **R2** may configure a second plane. Although FIG. 3 illustrates a case where the memory device **10** has a PUC structure, the disclosure is not limited thereto. As will be described later with reference to FIGS. 8 to 10, a memory device may have a POC (peri over cell) structure.

FIG. 4 is a top view illustrating a representation of a layout of a memory device in accordance with an embodiment of the disclosure.

Referring to FIG. 4, a memory device **10** may include a plurality of plane regions **R1** to **R4**. For instance, FIG. 4 illustrates a case including four plane regions, which are disposed in a line in the first direction **FD**. For the sake of convenience in explanation, the plane regions **R1** to **R4** will be defined as first to fourth plane regions **R1** to **R4**. Each of the first to fourth plane regions **R1** to **R4** may include a cell region **CR** and a row decoder region, **XR** which is adjacent to the cell region **CR** in the first direction **FD**.

A memory cell array **110**, a row decoder **120** and a page buffer circuit **130** may be disposed in each of the first to fourth plane regions **R1** to **R4**. The memory cell array **110**, the row decoder **120** and the page buffer circuit **130**, which are included in each of the first to fourth plane regions **R1** to **R4**, may configure one plane.

The memory cell array **110** may be disposed in the cell region **CR**. The row decoder **120** may be disposed in the row decoder region **XR**. The page buffer circuit **130** may be disposed in the cell region **CR** to overlap with the memory cell array **110** in the vertical direction **VD**.

As described above with reference to FIG. 2, operating voltages may be transferred to the word lines of the memory cell array **110** through the pass transistors of the row decoder **120**. In order to reduce a delay time caused in the process of transferring the operating voltages, the row decoder **120** may be disposed to have a shape extending in the second direction **SD**, as a direction in which the word lines are arranged, and may be disposed to have a length substantially the same as or similar to that of the memory cell array **110** in the second direction **SD**.

In each of the first to fourth plane regions **R1** to **R4**, the page buffer circuit **130** may be disposed by being divided into a first page buffer region (First PB) **131** and a second page buffer region (Second PB) **132**. The first page buffer

region **131** and the second page buffer region **132** may be arranged in the second direction SD.

In order to reduce a delay time of a signal, applied to bit lines by the page buffer circuit **130** or provided to the page buffer circuit **130** through the bit lines, the first page buffer region **131** and the second page buffer region **132** each may have a length substantially the same as or similar to that of the memory cell array **110** in the first direction FD, as a direction in which the bit lines are arranged.

A peripheral circuit may be disposed by being divided into a plurality of peripheral regions **210** and **220**. The peripheral regions **210** and **220** may include a plurality of first peripheral regions (Top PERI) **210** and a plurality of second peripheral regions (Middle PERI) **220**. One first peripheral region **210** and one second peripheral region **220** may be disposed in the cell region CR of each of the first to fourth plane regions R1 to R4 to overlap with the memory cell array **110** in the vertical direction VD.

In a single cell region CR, the second peripheral region **220** may be disposed between the first page buffer region **131** and the second page buffer region **132**. The first peripheral region **210** may be disposed on a side of the first page buffer region **131** opposite to the side adjacent to the second peripheral region **220**. Thus, the first peripheral region **210** and the second peripheral region **220**, which are disposed in the single cell region CR, may be positioned on both sides, respectively, of the first page buffer region **131**, which has a length substantially the same as or similar to that of the memory cell array **110** in the first direction FD.

The first peripheral region **210** of the first plane region R1 and the first peripheral region **210** of the second plane region R2 may be arranged in the first direction FD on opposite sides of the pair of the row decoder **120** of the first plane region R1 and the row decoder **120** of the second plane region R2. The row decoder **120** of the first plane region R1 and the row decoder **120** of the second plane region R2 may each have a length substantially the same as or similar to that of the memory cell array **110** in the second direction SD. The second peripheral region **220** of the first plane region R1 and the second peripheral region **220** of the second plane region R2 may also be positioned, in the first direction FD, on both sides of the pair of the row decoder **120** of the first plane region R1 and the row decoder **120** of the second plane region R2.

The first peripheral region **210** of the third plane region R3 and the first peripheral region **210** of the fourth plane region R4 may be arranged in the first direction FD on opposite sides of the pair of the row decoder **120** of the third plane region R3 and the row decoder **120** of the fourth plane region R4. The row decoder **120** of the third plane region R3 and the row decoder **120** of the fourth plane region R4 may each have a length substantially the same as or similar to that of the memory cell array **110** in the second direction SD. The second peripheral region **220** of the third plane region R3 and the second peripheral region **220** of the fourth plane region R4 may also be positioned, in the first direction FD, on both sides of the pair of the row decoder **120** of the third plane region R3 and the row decoder **120** of the fourth plane region R4.

Although not illustrated, elements that configure a peripheral circuit may be coupled to one another through coupling lines. The coupling lines may include power lines that serve to transmit power such as a power voltage and a ground voltage, analog signal lines that serve to transmit analog signals, and logic signal lines that serve to transmit logic signals such as data, commands and addresses.

In order to couple elements that configure the peripheral circuit disposed in the plurality of peripheral regions **210** and **220**, some of the coupling lines may have shapes extending in the second direction SD, and some of the coupling lines may have shapes extending in the first direction FD.

As described above with reference to FIG. 2, each of the row decoders **120** may include a plurality of pass transistors. The pass transistors may serve to transfer operating voltages, applied to global lines (GWL of FIG. 2), to the memory cell array **110**. Although not illustrated, the global lines may extend in the second direction SD, and may be disposed to overlap with the pass transistors of the row decoder **120** in the vertical direction VD.

As described above, because the global lines extending in the second direction SD are disposed over the pass transistors of the row decoders **120**, disposing the coupling lines to avoid the global lines becomes more difficult or complicated.

For example, if a plane area is increased or a wiring layer is added for the disposition of the coupling lines, it may be possible to dispose the coupling lines independent of the positions of the global lines. However, if a plane area is increased or a wiring layer is added, the size of the memory device **10** will increase.

Embodiments of the disclosure may enable the disposition of coupling lines without increasing the size of a memory device.

FIGS. 5 and 6 are cross-sectional views illustrating a representation of a memory device in accordance with an embodiment of the disclosure.

FIG. 5 is a cross-sectional view illustrating a cell region CR, and FIG. 6 is a cross-sectional view illustrating a row decoder region XR. Components indicated by dotted lines in FIG. 6 are illustrated to facilitate understanding, and are not actually disposed on the cross-section of FIG. 6.

Referring to FIGS. 5 and 6, a substrate **1** may include at least one selected from the group including a monocrystalline silicon layer, an SOI (silicon on insulator), a silicon layer formed on a silicon germanium (SiGe) layer, a monocrystalline silicon layer formed on a dielectric layer and a polysilicon layer formed on a dielectric layer.

A logic structure **2** may include a plurality of transistors TR, which are defined in active regions of the substrate **1** defined by an isolation layer **1A**, a plurality of bottom wiring layers UM1 to UM3, and a plurality of contacts UM1C to UM3C.

A transistor TR may include a gate dielectric layer Gox that is defined on the substrate **1**, a gate electrode G that is defined on the gate dielectric layer Gox, and junction regions, which are formed by implanting impurity ions into the active region on both sides of the gate electrode G. One of the junction regions may be used as a source region S of the transistor TR, and the other may be used as a drain region D of the transistor TR. The transistor TR may configure any one of a row decoder, a page buffer circuit and a peripheral circuit.

A dielectric layer **21** may be defined on the substrate **1** to cover the transistors TR. The dielectric layer **21** may include silicon oxide, for example, HDP (high density plasma) oxide or TEOS (tetra-ethyl-ortho-silicate) oxide.

The plurality of bottom wiring layers UM1 to UM3 may be disposed in the dielectric layer **21**. For example, the bottom wiring layers UM1 to UM3 may include a first bottom wiring layer UM1, a second bottom wiring layer UM2 over the first bottom wiring layer UM1, and a third bottom wiring layer UM3 over the second bottom wiring

layer UM2. While FIGS. 5 and 6 illustrate three bottom wiring layers UM1 to UM3, it is to be noted that the disclosure is not limited thereto. The number of wiring layers may be four or more.

A plurality of wiring line patterns may be disposed in each of the bottom wiring layers UM1 to UM3. The wiring line patterns of the bottom wiring layers UM1 to UM3 may be configured to have properties that may not exhibit a process failure, for example, a hillock, at a maximum temperature (hereinafter, referred to as a 'process critical temperature'), during a process of forming a memory cell array. In other words, as a material for the wiring line patterns of the bottom wiring layers UM1 to UM3, a conductive material that has a heat resistance characteristic at the process critical temperature may be used. For example, the wiring line patterns of the bottom wiring layers UM1 to UM3 may include a material that has a melting point higher than the process critical temperature, such as tungsten (W). Because the wiring line patterns of the bottom wiring layers UM1 to UM3 are formed before the memory cell array is formed, the wiring line patterns are formed of a conductive material that has a high resistivity and has a high melting point.

The wiring line patterns of the second bottom wiring layer UM2 may include a plurality of first local lines LWL1, a plurality of global lines GWL and a plurality of first coupling lines CL1. The wiring line patterns of the third bottom wiring layer UM3 may include a plurality of first local pads LP1 and a plurality of second coupling lines CL2 (not illustrated in detail). The plurality of first local lines LWL1, the plurality of global lines GWL, the plurality of first coupling lines CL1 and the plurality of first local pads LP1 may be disposed in the row decoder region XR. Although not illustrated in detail, the plurality of second coupling lines CL2 may traverse the row decoder region XR in the first direction FD, and an end of each of the plurality of second coupling lines CL2 may be disposed in the cell region CR.

The contact UM1C may be defined between the first bottom wiring layer UM1 and the transistor TR, and may couple a wiring line pattern of the first bottom wiring layer UM1 and the transistor TR. The contact UM2C may be defined between the second bottom wiring layer UM2 and the first bottom wiring layer UM1, and may couple a wiring line pattern of the second bottom wiring layer UM2 and a wiring line pattern of the first bottom wiring layer UM1. The contact UM3C may be defined between the third bottom wiring layer UM3 and the second bottom wiring layer UM2, and may couple a wiring line pattern of the third bottom wiring layer UM3 and a wiring line pattern of the second bottom wiring layer UM2.

A source plate 3 may be configured to be disposed on the dielectric layer 21 of the cell region CR and to expose the dielectric layer 21 of the row decoder region XR. The source plate 3 may be formed as a polysilicon layer. Unlike the substrate 1, which may use a monocrystalline silicon layer, the source plate 3 may be formed as a polysilicon layer on the dielectric layer 21 of the logic structure 2.

A memory structure 4 may include a plurality of electrode layers 41 and a plurality of interlayer dielectric layers 42, which are alternately stacked on the source plate 3, a plurality of vertical channels CH, which pass through the plurality of electrode layers 41 and the plurality of interlayer dielectric layers 42, a plurality of top wiring layers M1 and M2, and contacts M0C, M1C and M2C.

The electrode layers 41 may include a conductive material. For example, the electrode layers 41 may include at least one selected among a doped semiconductor (e.g., doped silicon), a metal (e.g., tungsten, copper or aluminum),

a conductive metal nitride (e.g., titanium nitride or tantalum nitride) and a transition metal (e.g., titanium or tantalum). Among the electrode layers 41, at least one electrode layer 41 from the lowermost electrode layers 41 may configure a source select line SSL. Among the electrode layers 41, at least one electrode layer 41 from the uppermost electrode layers 41 may configure a drain select line DSL. The electrode layers 41 between the source select line SSL and the drain select line DSL may configure word lines WL.

Each of the vertical channels CH may include a channel layer and a gate dielectric layer. The channel layer may include polysilicon or monocrystalline silicon, and may include, in some regions thereof, a p-type impurity such as boron (B). The channel layer may have the shape of a pillar that is completely filled up to its center, or a solid cylinder. While not illustrated, the channel layer may have the shape of a tube whose center region is open. In this case, a buried dielectric layer may be formed in the open center region of the channel layer. The gate dielectric layer may have the shape of a straw or a cylindrical shell that surrounds the outer wall of the channel layer. Although not illustrated, the gate dielectric layer may include a tunnel dielectric layer, a charge storage layer and a blocking layer, which are sequentially stacked from the outer wall of the channel layer. The gate dielectric layer may have an ONO (oxide-nitride-oxide) stack structure in which an oxide layer, a nitride layer and an oxide layer are sequentially stacked.

A source select transistor may be formed in areas or regions where the source select line SSL surrounds the vertical channel CH. A drain select transistor may be formed in areas or regions where the drain select line DSL surrounds the vertical channel CH. A memory cell may be formed in areas or regions where the word line WL surrounds the vertical channel CH.

A dielectric layer 43 may be defined on the logic structure 2 to cover the source plate 3, the top surfaces and side surfaces of the plurality of electrode layers 41 and the plurality of interlayer dielectric layers 42, which are alternately stacked, and the side surfaces of the plurality of vertical channels CH. In the row decoder region XR, a of contacts M0C pass through the dielectric layer 43 in the vertical direction VD and may be coupled respectively to the plurality of first local pads LP1 and to at least one second coupling line CL2.

A dielectric layer 44 may be defined on the dielectric layer 43. The plurality of top wiring layers M1 and M2 may be disposed in the dielectric layer 44. For example, the top wiring layers M1 and M2 may include a first top wiring layer M1 and a second top wiring layer M2 over the first top wiring layer M1. Each of the top wiring layers M1 and M2 may include a plurality of wiring line patterns.

The wiring line patterns of the top wiring layers M1 and M2 may be formed of a conductive material that has a lower resistivity than the wiring line patterns of the bottom wiring layers UM1 to UM3. For example, the wiring line patterns of the top wiring layers M1 and M2 may include at least one of aluminum (Al) and copper (Cu). Because the wiring line patterns of the top wiring layers M1 and M2 are formed after the memory cell array is formed, they may be formed of a material that may cause a process failure at the process critical temperature due to a low melting point thereof and that has a low resistivity.

Contacts M1C may be defined on the contacts M0C and the vertical channels CH. The wiring line patterns of the first top wiring layer M1 may include bit lines BL, second local lines LWL2 and a coupling pad CP1. The bit lines BL may be disposed in the cell region CR, extend in the second

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direction SD, and arranged in the first direction FD. Each of the bit lines BL may be coupled to a corresponding vertical channel CH through the contact M1C.

The second local lines LWL2 and the coupling pad CP1 may be disposed in the row decoder region XR. Each of the second local lines LWL2 and the coupling pad CP1 may be coupled to a corresponding contact M0C through a contact M1C.

A plurality of contacts M2C may be defined between the first top wiring layer M1 and the second top wiring layer M2. Each of the plurality of contacts M2C may couple a wiring line pattern of the first top wiring layer M1 and a corresponding wiring line pattern of the second top wiring layer M2. The second top wiring layer M2 may include a third coupling line CL3. The third coupling line CL3 may be coupled to the coupling pad CP1 through the contact M2C.

FIGS. 7A to 7E are top views illustrating representations of main components of a memory device in accordance with an embodiment of the disclosure, and illustrate the row decoder region XR.

In detail, FIG. 7A is a top view illustrating pass transistors TR of a row decoder; FIG. 7B is a top view illustrating pass transistors TR and a second bottom wiring layer UM2; FIG. 7C is a top view illustrating a second bottom wiring layer UM2, a third bottom wiring layer UM3 and contacts UM3C; FIG. 7D is a top view illustrating a third bottom wiring layer UM3, a first top wiring layer M1 and contacts M0C; and FIG. 7E is a top view illustrating a first top wiring layer M1, a second top wiring layer M2 and contact M2C.

Referring to FIGS. 6 and 7A, a plurality of active regions ACT may be formed in the row decoder region XR of the substrate 1. In each of the active regions ACT, one drain region D and two source regions S, which are doped into a first conductivity type, and two channel regions (not illustrated) between the drain region D and source regions S, which are doped into a second conductivity type may be defined. The drain region D may be disposed, in the second direction SD, at a center portion of the active region ACT. The two source regions S may be disposed at both end portions, respectively, of the active region ACT in the second direction SD. Each of the channel regions may be disposed between a drain region D and an adjacent source region S. The gate electrode G may be formed over each channel region with the gate dielectric layer Gox interposed therebetween. Gate electrodes G may include a conductive material such as polysilicon. Two pass transistors TR that share one drain region D may be configured in each of the active regions ACT. The two pass transistors TR that share one drain region D may be coupled to different memory blocks BLK, respectively. Pass transistors TR of different active regions ACT, which are coupled to one memory block BLK, may be disposed in two rows that are adjacent to each other in the second direction SD.

Although the present embodiment illustrates two pass transistors TR in one active region ACT, it is to be noted that the disclosure is not limited thereto. For example, one pass transistor TR may be configured in one active region ACT.

Although the present embodiment illustrates pass transistors TR, which are coupled to one memory block BLK, disposed in two rows, it is to be noted that the disclosure is not limited thereto. Pass transistors TR that are coupled to one memory block BLK may be disposed in one row or at least three rows, for example.

Referring to FIGS. 6 and 7B, the plurality of first local lines LWL1, the plurality of global lines GWL and the

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plurality of first coupling lines CL1 may be disposed in the second bottom wiring layer UM2 of the row decoder region XR.

A plurality of wiring line tracks T, which extend in the second direction SD and are arranged at regular intervals in the first direction FD, may be defined in the second bottom wiring layer UM2. The plurality of wiring line tracks T may be lines that serve as references in disposing the first local lines LWL1, the global lines GWL and the first coupling lines CL1. The first local lines LWL1, the global lines GWL and the first coupling lines CL1 may be aligned with the wiring line tracks T, and the center line of each of the first local lines LWL1, the global lines GWL and the first coupling lines CL1 may overlap with a corresponding wiring line track T.

The global lines GWL may extend in the second direction SD, and may each overlap with corresponding pass transistors TR in the vertical direction VD. Each of the plurality of global lines GWL may be coupled to the drain regions D (see FIG. 7A) of the corresponding pass transistors TR through the contact UM2C (see FIG. 6), the wiring line pattern of the first bottom wiring layer UM1 (see FIG. 6) and the contact UM1C (see FIG. 6). The second bottom wiring layer UM2 may correspond to an uppermost wiring layer that is used in disposing global lines. Although the present embodiment illustrates global lines disposed in one bottom wiring layer, it is to be noted that the disclosure is not limited thereto. The global lines may be disposed in at least one of remaining bottom wiring layers, except for the third bottom wiring layer UM3.

The first local lines LWL1 may extend in the second direction SD, and may each overlap with the source region S of a corresponding pass transistor TR in the vertical direction VD. Each of the plurality of first local lines LWL1 may be coupled to the source region S (see FIG. 6) of a corresponding pass transistor TR through the contact UM2C (see FIG. 6), the wiring line pattern of the first bottom wiring layer UM1 (see FIG. 6), and the contact UM1C (see FIG. 6).

Each of the plurality of first local lines LWL1 may be disposed within a width of a corresponding memory block BLK in the second direction SD. A plurality of first local lines LWL1 corresponding to different memory blocks BLK may be disposed in one wiring line track T. That is to say, a plurality of first local lines LWL1 corresponding to different memory blocks BLK may share one wiring line track T.

In the second bottom wiring layer UM2, there may exist wiring line tracks T that are not used for disposing the global lines GWL and the first local lines LWL1. A region AR of FIG. 7B may correspond to a region where wiring line tracks T are positioned that are not used for disposing the global lines GWL and the first local lines LWL1. The wiring line tracks T of the region AR may be used for the first coupling lines CL1.

Referring to FIGS. 6 and 7C, a plurality of second coupling lines CL2 and a plurality of first local pads LP1 may be disposed in the third bottom wiring layer UM3. The first local pads LP1 corresponding to one memory block BLK may be disposed in one row within the width of the corresponding memory block BLK. The plurality of first local pads LP1 may be disposed in a plurality of rows in the second direction SD corresponding to the plurality of memory blocks BLK disposed in the second direction SD. The first local pads LP1 may be coupled to the corresponding first local lines LWL1 through the contact UM3C (see FIG. 6).

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A first zone ZONE1 and a second zone ZONE2, which are separated by a row of the first local pads LP1 may be defined. The first zone ZONE1 and the second zone ZONE2 may be disposed in the second direction SD. Although not illustrated, a plurality of first zones ZONE1 and a plurality of second zones ZONE2 may be alternately arranged in the second direction SD.

The second coupling lines CL2 may be disposed in the first zone ZONE1 and the second zone ZONE2, and may have line shapes that extend in the first direction FD. In the second direction SD, a width of the second coupling lines CL2 disposed in the first zone ZONE1 may be larger than a width of the second coupling lines CL2 disposed in the second zone ZONE2.

Due to the resistance components of the second coupling lines CL2, the magnitudes of signals may decrease during a process in which the signals are transferred through the second coupling lines CL2. The second coupling lines CL2 of the first zone ZONE1 may be used to transmit a signal that has a high possibility of causing a functional error of the memory device when the magnitude thereof decreases, for example, power and analog signals. The second coupling lines CL2 of the second zone ZONE2 may be used to transmit a signal that has a low possibility of causing a functional error of the memory device even when the magnitude thereof decreases and requires a large number of lines, for example, a logic signal that only needs to be distinguished between a logic high and a logic low.

It is not necessary to configure pads that are coupled to the global lines GWL in the third bottom wiring layer UM3 because the second bottom wiring layer UM2 is an uppermost wiring layer that is used in disposing global lines. For this reason, no pads that are coupled to the global lines GWL exist in the third bottom wiring layer UM3.

Unlike the present embodiment, in cases where pads that are coupled to global lines exist in a third bottom wiring layer, the number of pads disposed in the third bottom wiring layer increases, and the number of rows of the pads increases. Thus, the number of second coupling lines capable of being disposed in the third bottom wiring layer may decrease. In contrast, according to the present embodiment, no pads that are coupled to the global lines GWL exist in the third bottom wiring layer UM3, so the number of pads disposed in the third bottom wiring layer UM3 may be reduced, and the number of rows of the pads may be reduced. Thus, the number of second coupling lines capable of being disposed in the third bottom wiring layer UM3 may be increased.

Referring to FIGS. 6 and 7D, a plurality of second local lines LWL2 and a coupling pad CP1 may be disposed in the first top wiring layer M1.

The second local lines LWL2 may extend in the second direction SD, and may each overlap with a corresponding first local pad LP1 in the vertical direction VD. The contact M1C (see FIG. 6) and the contact M0C may be disposed in the vertical direction VD at an overlapping region between the second local line LWL2 and the corresponding first local pad LP1, and, thereby, may couple the second local line LWL2 and the first local pad LP1.

The coupling pad CP1 may overlap with one of the second coupling lines CL2 of the first zone ZONE1 in the vertical direction VD. The contact M1C (see FIG. 6) and the contact M0C may be disposed at an overlapping region between the coupling pad CP1 and the second coupling line CL2, and thereby, may couple the coupling pad CP1 and the second coupling line CL2.

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Referring to FIGS. 6 and 7E, a third coupling line CL3 may be disposed in a second top wiring layer M2. The third coupling line CL3 may extend in the second direction SD, and may overlap with the coupling pad CP1 in the vertical direction VD. The contact M2C may be defined at an overlapping region between the third coupling line CL3 and the coupling pad CP1, and the third coupling line CL3 may be coupled to the coupling pad CP1 through the contact M2C.

The third coupling line CL3 may be coupled to one of the second coupling lines CL2 of the first zone ZONE1 (see FIG. 7D) through the contact M2C, the coupling pad CP1, the contact M1C and the contact M0C.

Like the second coupling lines CL2 (see FIG. 7D) of the first zone ZONE1 (see FIG. 7D), the third coupling line CL3 may be used to transmit a signal that has a high possibility of causing a functional error of the memory device when the magnitude thereof decreases, for example, power and analog signals. As described above, the top wiring layer M2 is made of a conductive material that has a lower resistivity than the bottom wiring layers UM1 to UM3. By using the third coupling line CL3 to transmit a signal, such as a power and an analog signal, which is likely to cause a functional error of the memory device when the level thereof decreases, it is possible to suppress the occurrence of a functional error of the memory device.

No global lines are disposed in the first top wiring layer M1 and the second top wiring layer M2. The first top wiring layer M1 and the second top wiring layer M2 may be defined as layers in which the disposition of global lines is prohibited.

A high voltage may be loaded on the second local lines LWL2 depending on a memory operation state. The high voltage applied to the second local lines LWL2 may be higher than a voltage applied to the third coupling line CL3.

Because voltage driving ranges of the second local lines LWL2 and the third coupling line CL3 are different, the memory device may malfunction due to the coupling between the second local lines LWL2 and the third coupling line CL3. According to the present embodiment, because the disposition of global lines in the first top wiring layer M1 and the second top wiring layer M2 is prohibited, it is possible to reduce the number of top wiring layers required for the disposition of the second local lines LWL2. Accordingly, it is possible to prohibit local lines from being disposed in the second top wiring layer M2 in which the third coupling line CL3 is disposed, and it is possible to prohibit coupling lines from being disposed in the first top wiring layer M1 in which the second local lines LWL2 are disposed.

Therefore, it is possible to dispose the third coupling lines CL3 and the second local lines LWL2 in different top wiring layers. Accordingly, the coupling between coupling lines and local lines may be reduced, thereby contributing to suppressing a malfunction of the memory device due to coupling.

FIGS. 8 to 10 are cross-sectional views of a memory device in accordance with an embodiment of the disclosure. In detail, FIG. 8 is a cross-sectional view taken by cutting a part including first and second local pads LP1 and LP2 in a row decoder region XR, FIG. 9 is a cross-sectional view taken by cutting a part including second and third coupling lines CL2 and CL3 in the row decoder region XR, and FIG. 10 is a cross-sectional view of a cell region CR.

Referring to FIGS. 8 to 10, the memory device in accordance with the embodiment of the disclosure may include a first semiconductor structure 100 and a second semiconductor

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tor structure **200**. The second semiconductor structure **200** may include a memory cell array, and the first semiconductor structure **100** may include a logic circuit which controls the operation of the memory cell array.

The first semiconductor structure **100** and the second semiconductor structure **200** may be fabricated on different wafers and be then coupled by being bonded to each other. In this case, the memory device may be defined as having a POC (peri over cell) structure.

The first semiconductor structure **100** may include a substrate **1**, a plurality of transistors **TR1** and **TR2** which are defined on the substrate **1**, a first bonding layer **BNL1**, and a plurality of bottom wiring layers **UM1** to **UM5** which are disposed between the plurality of transistors **TR1** and **TR2** and the first bonding layer **BNL1**.

The plurality of transistors **TR1** and **TR2** may include first transistors **TR1** which are disposed in the row decoder region **XR** of the substrate **1** and second transistors **TR2** which are disposed in the cell region **CR** of the substrate **1**. The first transistor **TR1** may be the pass transistor of a row decoder **120**. In order to simplify explanation and facilitate understanding, hereinafter, a pass transistor will be denoted by the same reference numeral as the first transistor **TR1**. The second transistor **TR2** may constitute a page buffer. Although not illustrated in detail, a logic circuit including the row decoder **120**, a page buffer circuit **130** and a peripheral circuit (not illustrated) may be disposed in the substrate **1**.

A dielectric layer **21** may be defined on the substrate **1** to cover the logic circuit including the first and second transistors **TR1** and **TR2**.

The plurality of bottom wiring layers **UM1** to **UM5** may be disposed in the dielectric layer **21**. The bottom wiring layers **UM1** to **UM5** may include a first bottom wiring layer **UM1**, a second bottom wiring layer **UM2** over the first bottom wiring layer **UM1**, a third bottom wiring layer **UM3** over the second bottom wiring layer **UM2**, a fourth bottom wiring layer **UM4** over the third bottom wiring layer **UM3**, and a fifth bottom wiring layer **UM5** over the fourth bottom wiring layer **UM4**. The first bottom wiring layer **UM1** may be a lowermost layer which is closest to the substrate **1** among the bottom wiring layers **UM1** to **UM5**. Although FIGS. **8** to **10** illustrate a case including five bottom wiring layers **UM1** to **UM5**, it is to be noted that the disclosure is not limited thereto.

The wiring line patterns of the bottom wiring layers **UM1** to **UM5** may include, for example, at least one of aluminum (Al) and copper (Cu). Because the wiring line patterns of the bottom wiring layers **UM1** to **UM5** are formed on a wafer separate from the memory cell array, a material constituting the wiring line patterns of the bottom wiring layers **UM1** to **UM5** may be selected without considering the thermal budget of a process of forming the memory cell array. As a material constituting the wiring line patterns of the bottom wiring layers **UM1** to **UM5**, aluminum (Al), copper (Cu) or the like having a low resistivity may be selected. By this fact, the resistance of the wiring line patterns of the bottom wiring layers **UM1** to **UM5** may be reduced.

The bottom wiring layers **UM1** to **UM5** may be divided into a first tier and a second tier. Bottom wiring layers of the first tier may be disposed at a bottom part, and bottom wiring layers of the second tier may be disposed at a top part. In the present embodiment, the first and second bottom wiring layers **UM1** and **UM2** may be included in the first tier, and the third to fifth bottom wiring layers **UM3** to **UM5** may be included in the second tier. Although FIGS. **8** to **10** illustrate a case where the number of bottom wiring layers of the first

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tier is two and the number of bottom wiring layers of the second tier is three, the disclosure is not limited thereto. Each of the number of bottom wiring layers of the first tier and the number of bottom wiring layers of the second tier may be two or more.

A plurality of global lines **GWL** may be disposed in the row decoder region **XR** of at least one of the bottom wiring layers **UM1** and **UM2** of the first tier. In the present embodiment, the plurality of global lines **GWL** may be disposed in the second bottom wiring layer **UM2**. The second bottom wiring layer **UM2** may be an uppermost layer in which the global lines **GWL** are disposed. The global lines **GWL** may not be disposed in the bottom wiring layers of the second tier, that is, the third to fifth bottom wiring layers **UM3** to **UM5**.

A plurality of first local lines **LWL1** and a plurality of first coupling lines **CL1** may be disposed in the row decoder region **XR** of the second bottom wiring layer **UM2**. A plurality of second coupling lines **CL2** and a plurality of first coupling pads **LP1** may be disposed in the row decoder region **XR** of the third bottom wiring layer **UM3**. A plurality of third coupling lines **CL3** and a plurality of second coupling pads **LP2** may be disposed in the row decoder region **XR** of the fourth bottom wiring layer **UM4**. A plurality of fourth coupling lines **CL4** and a plurality of third coupling pads **LP3** may be disposed in the row decoder region **XR** of the fifth bottom wiring layer **UM5**. Although the present embodiment illustrates a case where coupling lines are disposed in each of the bottom wiring layers **UM3** to **UM5** of the second tier, the disclosure is not limited thereto. Coupling lines may be disposed in at least one among the bottom wiring layers **UM3** to **UM5** of the second tier.

Contacts **UM1C** may be defined between the first bottom wiring layer **UM1** and the transistors **TR1** or **TR2** to couple the wiring line patterns of the first bottom wiring layer **UM1** and the transistors **TR1** or **TR2**. Contacts **UM2C** may be defined between the second bottom wiring layer **UM2** and the first bottom wiring layer **UM1** to couple the wiring line patterns of the second bottom wiring layer **UM2** and the wiring line patterns of the first bottom wiring layer **UM1**. Contacts **UM3C** may be defined between the third bottom wiring layer **UM3** and the second bottom wiring layer **UM2** to couple the wiring line patterns of the third bottom wiring layer **UM3** and the wiring line patterns of the second bottom wiring layer **UM2**. Contacts **UM4C** may be defined between the fourth bottom wiring layer **UM4** and the third bottom wiring layer **UM3** to couple the wiring line patterns of the fourth bottom wiring layer **UM4** and the wiring line patterns of the third bottom wiring layer **UM3**. Contacts **UM5C** may be defined between the fifth bottom wiring layer **UM5** and the fourth bottom wiring layer **UM4** to couple the wiring line patterns of the fifth bottom wiring layer **UM5** and the wiring line patterns of the fourth bottom wiring layer **UM4**.

The first local line **LWL1** may be coupled to the pass transistor **TR1** through the contact **UM2C**, the wiring line pattern of the first bottom wiring layer **UM1**, and the contact **UM21**. The first local pad **LP1** may be coupled to the first local line **LWL1** through the contact **UM3C**. The second local pad **LP2** may be coupled to the first local pad **LP1** through the contact **UM4C**. The third local pad **LP3** may be coupled to the second local pad **LP2** through the contact **UM5C**.

The first bonding layer **BNL1** may include a plurality of first bonding pads **PBD** and a first bonding insulating pattern **22** which insulates the plurality of first bonding pads **PBD** from each other. First bonding contacts **PBC** may be defined

between the first bonding layer BNL1 and the fifth bottom wiring layer UM5 to couple the first bonding pads PBD and the wiring line patterns of the fifth bottom wiring layer UM5. The first bonding pad PBD may be coupled to the third local pad LP3 through the first bonding contact PBC.

The second semiconductor structure 200 may include a second bonding layer BNL2, the memory cell array, and at least one top wiring layer M1 which is disposed between the second bonding layer BNL2 and the memory cell array. Although the present embodiment illustrates a case where one top wiring layer is disposed between the second bonding layer BNL2 and the memory cell array, the disclosure is not limited thereto. The number of top wiring layers disposed between the second bonding layer BNL2 and the memory cell array may be at least one. For the sake of convenience in explanation, hereinafter, a top wiring layer disposed between the second bonding layer BNL2 and the memory cell array will be defined as a first top wiring layer M1.

The second bonding layer BNL2 may include a plurality of second bonding pads CBD and a second bonding insulating pattern 43A.

The second bonding layer BNL2 of the second semiconductor structure 200 may be bonded to the first bonding layer BNL1 of the first semiconductor structure 100 in a face-to-face manner at a bonding interface BS. The bonding interface BS may be configured between the first bonding layer BNL1 and the second bonding layers BNL2, as a result of hybrid bonding that is also referred to as direct bonding and may simultaneously configure a metal-metal bond and a dielectric-dielectric bond. At the bonding interface BS, the plurality of first bonding pads PBD and the plurality of second bonding pads CBD may be bonded to each other to configure a plurality of conductive bonds, and the first bonding insulating pattern 22 and the second bonding insulating pattern 43A may be bonded to each other to configure an insulating bond.

A dielectric layer 43B may be disposed on the second bonding layer BNL2. The first top wiring layer M1 may be disposed in the dielectric layer 43B. A plurality of second local lines LWL2 and a plurality of bit lines BL may be configured in the first top wiring layer M1. The second local lines LWL2 may be disposed in the row decoder region XR of the first top wiring layer M1, and the bit lines BL may be disposed in the cell region CR of the first top wiring layer M1.

A dielectric layer 43C may be disposed on the dielectric layer 43B. A stack STA including a plurality of electrode layers 41 and a plurality of interlayer dielectric layers 42 which are alternately stacked may be disposed on the dielectric layer 43C. The plurality of electrode layers 41 may include a plurality of word lines and at least one select line. A stairway structure may be defined in the row decoder region XR of the stack STA. The dielectric layer 43C may fill the stairway structure of the stack STA.

A plurality of vertical channels CH which pass through the cell region CR of the stack STA may be provided. Memory cells may be configured in regions where the word lines surround the vertical channels CH. Select transistors may be configured in regions where the select line surrounds the vertical channels CH. The memory cells and the select transistors may configure the memory cell array.

Word line contacts WCT may be configured in the dielectric layer 43C. The word line contacts WCT may be coupled to the plurality of electrode layers 41 in the row decoder region XR.

Bit line contacts BCT may be defined in the cell region CR of the dielectric layer 43B to couple the plurality of

vertical channels CH and the bit lines BL. Contacts M1C may be defined in the row decoder region XR of the dielectric layer 43B to couple the word line contacts WCT and the second local lines LWL2.

A plurality of second bonding contacts CBC may be defined between the first top wiring layer M1 and the second bonding layer BNL2 to couple the second local lines LWL2 and the bit lines BL of the first top wiring layer M1 and the second bonding pads CBD of the second bonding layer BNL2.

A source plate 3 may be disposed on the cell region CR of the stack STA. The upper ends of the vertical channels CH may be coupled to the source plate 3. A dielectric layer 43D may be disposed on the row decoder region XR of the stack STA. A dielectric layer 44A may be disposed on the source plate 3 and the dielectric layer 43D, and a second top wiring layer M2 may be disposed in the dielectric layer 44A.

FIGS. 11A to 11E are top views illustrating representations of main components of a row decoder region XR of a memory device in accordance with an embodiment of the disclosure.

In detail, FIG. 11A is a top view illustrating pass transistors TR1, FIG. 11B is a top view illustrating the pass transistors TR1 and a second bottom wiring layer UM2, FIG. 11C is a top view illustrating the second bottom wiring layer UM2 and a third bottom wiring layer UM3, FIG. 11D is a top view illustrating a fourth bottom wiring layer UM4, and FIG. 11E is a top view illustrating the fourth bottom wiring layer UM4 and a fifth bottom wiring layer UM5.

Referring to FIG. 11A, a plurality of active regions ACT may be provided in the row decoder region XR of a substrate in the first direction FD and the second direction SD. Two pass transistors TR1 which share one drain region D may be configured in each of the plurality of active regions ACT. Two pass transistors TR1 which share one drain region D may be coupled to different memory blocks BLK, respectively. Although not illustrated in detail, pass transistors TR1 which are coupled to four memory blocks BLK may be disposed in six rows. FIG. 11A illustrates four pass transistor rows among six pass transistor rows which are coupled to four memory blocks BLK. Although the present embodiment illustrates a case where pass transistors TR1 which are coupled to four memory blocks BLK are disposed in six rows, it is to be noted that the disclosure is not limited thereto.

Referring to FIGS. 8, 9 and 11B, a plurality of global lines GWL, a plurality of first local lines LWL1 and a plurality of first coupling lines CL1 may be disposed in the row decoder region XR of the second bottom wiring layer UM2.

Each of the global lines GWL may have a line shape which extends in the second direction SD. Each of the plurality of global lines GWL may be coupled to the drain region D (see FIG. 11A) of a corresponding pass transistor TR1 through a contact (UM2C of FIG. 8), the wiring line pattern of a first bottom wiring layer (UM1 of FIG. 8) and a contact (UM1C of FIG. 8). The second bottom wiring layer UM2 may be an uppermost layer which is used in disposing the global lines GWL. Although the present embodiment illustrates a case where global lines are disposed in one bottom wiring layer, it is to be noted that the disclosure is not limited thereto. Global lines may be disposed in at least one of the bottom wiring layers UM1 and UM2 of a first tier.

Each of the plurality of first local lines LWL1 may be coupled to a source region S (see FIG. 11A) of a corresponding pass transistor TR1 through a contact (UM2C of FIG. 8), the wiring line pattern of the first bottom wiring layer (UM1 of FIG. 8) and a contact (UM1C of FIG. 8).

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The plurality of first coupling lines CL1 may be disposed in a region which is not used in disposing the plurality of global lines GWL and the plurality of first local lines LWL1. Each of the plurality of first coupling lines CL1 may have a line shape which extends in the second direction SD being the extending direction of the plurality of global lines GWL. In order to prevent a failure in a process due to a difference in pattern density, dummy patterns DUMMY may be disposed in regions in which the plurality of global lines GWL, the plurality of first local lines LWL1 and the plurality of first coupling lines CL1 are not disposed.

Referring to FIGS. 8, 9 and 11C, a plurality of first local pads LP1 and a plurality of second coupling lines CL2 may be disposed in the row decoder region XR of the third bottom wiring layer UM3.

The first local pads LP1 may be disposed in a plurality of rows in the second direction SD.

A plurality of first zones ZONE1 and a plurality of second zones ZONE2 which are divided by the plurality of rows of the first local pads LP1 may be defined in the third bottom wiring layer UM3. The plurality of first zones ZONE1 and the plurality of second zones ZONE2 may be alternately disposed two by two in the second direction SD. Although not illustrated, the plurality of first zones ZONE1 and the plurality of second zones ZONE2 may be alternately disposed one by one in the second direction SD.

The second coupling lines CL2 may be disposed in each of the first zone ZONE1 and the second zone ZONE2. Each of the second coupling lines CL2 may have a line shape which extends in the first direction FD. The second coupling lines CL2 of the first zone ZONE1 and the second coupling lines CL2 of the second zone ZONE2 may be used to transmit signals which have different characteristics. By this fact, interference between signals which have different characteristics may decrease, and thus, a failure of the memory device due to interference between signals which have different characteristics may be suppressed or prevented.

The second coupling lines CL2 of the first zone ZONE1 and the second coupling lines CL2 of the second zone ZONE2 may have different widths. For example, the width of the second coupling lines CL2 of the second zone ZONE2 may be smaller than the width of the second coupling lines CL2 of the first zone ZONE1. In this case, the second coupling lines CL2 of the first zone ZONE1 may be used to transmit a signal which has a high possibility of causing a functional error of the memory device when the magnitude of the signal decreases, for example, power and an analog signal. The second coupling lines CL2 of the second zone ZONE2 may be used to transmit a signal which has a low possibility of causing a functional error of the memory device even when the magnitude of the signal decreases, for example, a digital signal.

Referring to FIGS. 8, 9 and 11D, a plurality of second local pads LP2 and a plurality of third coupling lines CL3 may be disposed in the row decoder region XR of the fourth bottom wiring layer UM4.

The plurality of second local pads LP2 may have the same layout structure as the plurality of first local pads LP1, but the disclosure not limited thereto. The plurality of third coupling lines CL3 may have the same layout structure as the plurality of second coupling lines CL2, but the disclosure is not limited thereto.

The plurality of second local pads LP2 may be disposed in a plurality of rows in the second direction SD. A plurality of first zones ZONE1 and a plurality of second zones ZONE2 which are divided by the plurality of rows of the second local pads LP2 may be defined in the fourth bottom

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wiring layer UM4. The third coupling lines CL3 may be disposed in the first zones ZONE1 and the second zones ZONE2. Each of the third coupling lines CL3 may have a line shape which extends in the first direction FD. The third coupling lines CL3 of the first zone ZONE1 and the third coupling lines CL3 of the second zone ZONE2 may be used to transmit signals which have different characteristics. By this fact, a failure of the memory device due to interference between signals which have different characteristics may decrease.

The third coupling lines CL3 of the first zone ZONE1 and the third coupling lines CL3 of the second zone ZONE2 may have different widths. For example, the width of the third coupling lines CL3 of the second zone ZONE2 may be smaller than the width of the third coupling lines CL3 of the first zone ZONE1. In this case, the third coupling lines CL3 of the first zone ZONE1 may be used to transmit a signal which has a high possibility of causing a functional error of the memory device when the magnitude of the signal decreases, for example, power and an analog signal. The third coupling lines CL3 of the second zone ZONE2 may be used to transmit a signal which has a low possibility of causing a functional error of the memory device even when the magnitude of the signal decreases, for example, a digital signal.

Referring to FIGS. 8, 9 and 11E, a plurality of fourth coupling lines CL4 and a plurality of third local pads LP3 may be disposed in the row decoder region XR of the fifth bottom wiring layer UM5.

The plurality of fourth coupling lines CL4 may extend in the second direction SD, and may be arranged in the first direction FD. When viewed from the top, the fourth coupling lines CL4 may cross the third coupling lines CL3 and the second local pads LP2 of the fourth bottom wiring layer UM4. The fourth coupling line CL4 and the third coupling line CL3 may overlap with each other at an intersection. A contact UM5C may be disposed at the intersection between the fourth coupling line CL4 and the third coupling line CL3 to couple the fourth coupling line CL4 and the third coupling line CL3.

The third local pad LP3 may be disposed to overlap with a corresponding second local pad LP2. In correspondence to the layout structure of the second local pads LP2 which are disposed in a plurality of rows in the second direction SD, the third local pads LP3 may be disposed in a plurality of rows in the second direction SD.

The third local pads LP3 which are disposed in one row may be disposed alternately with the fourth coupling lines CL4 in the first direction FD. Two third local pads LP3 which are disposed adjacent to each other in each row and one fourth coupling line CL4 which is positioned therebetween may cross one second local pad LP2. Contacts (UM5C of FIG. 8) may be disposed at intersections where the second local pad LP2 and the third local pads LP3 intersect with each other, to couple the second local pad LP2 and the third local pads LP3. Accordingly, one second local pad LP2 may be coupled to two third local pads LP3, and may be coupled to two first bonding pads (PBD of FIG. 8) through two first bonding contacts (PBC of FIG. 8) which are coupled to the two third local pads LP3.

Since, as described above, the second bottom wiring layer UM2 is an uppermost layer which is used in disposing global lines, it is not necessary to configure, in the third to fifth bottom wiring layers UM3 to UM5, pads which are coupled to the global lines GWL. For this reason, no pads which are coupled to the global lines GWL exist in the third to fifth bottom wiring layers UM3 to UM5.

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Unlike the present embodiment, when pads which are coupled to global lines exist in third to fifth bottom wiring layers, an area is consumed due to the disposition of the pads, and thus, the number of coupling lines which can be disposed in the third to fifth bottom wiring layers will decrease. According to the present embodiment, since pads which are coupled to the global lines GWL do not exist in the third to fifth bottom wiring layers UM3 to UM5, it is possible to reduce the number of pads disposed in the third to fifth bottom wiring layers UM3 to UM5 and increase an area to be used in disposing coupling lines. Accordingly, a larger number of coupling lines may be disposed in the third to fifth bottom wiring layers UM3 to UM5, and it is not necessary to increase a layout area for disposing coupling lines, whereby it is possible to achieve high integration.

Referring back to FIG. 8, the plurality of second local lines LWL2 may be disposed in the first top wiring layer M1. The second local line LWL2 may be coupled to the electrode layer 41 through the contact M1C and the word line contact WCT. The second local line LWL2 may be coupled to the second bonding pad CBD through the second bonding contact CBC.

One second local line LWL2 may be coupled to two second bonding contacts CBC, and may be coupled to two second bonding pads CBD through the two second bonding contacts CBC.

Since, as described above, one second local pad LP2 is coupled to two first bonding pads PBD and one second local line LWL2 is coupled to two second bonding pads CBD, two conductive bonds which couple one second local pad LP2 and one second local line LWL2 may be configured. Accordingly, even when a failure in which the first bonding pad PBD and the second bonding pad CBD are not coupled occurs in any one of the two conductive bonds, the second local pad LP2 and the second local line LWL2 may be coupled through the other normal conductive bond. Accordingly, since a bonding failure does not lead to a failure of the memory device, the yield may be improved.

FIG. 12 is a block diagram schematically illustrating a representation of a memory system including a memory device in accordance with an embodiment of the disclosure.

Referring to FIG. 12, a memory system 600 in accordance with an embodiment may include a nonvolatile memory device (NVM Device) 610 and a memory controller 620.

The nonvolatile memory device (NVM Device) 610 may be constituted by a memory device described above and may operate in the manner described above. The memory controller 620 may be configured to control the nonvolatile memory device (NVM Device) 610. By the combination of the nonvolatile memory device (NVM Device) 610 and the memory controller 620, a memory card or a solid state disk (SSD) may be provided. An SRAM 621 is used as a working memory of a processing unit (CPU) 622. A host interface (Host I/F) 623 includes a data exchange protocol of a host, which is coupled with the memory system 600.

An error correction code block (ECC) 624 detects and corrects an error included in data read from the nonvolatile memory device (NVM Device) 610.

A memory interface (Memory I/F) 625 interfaces with the nonvolatile memory device (NVM Device) 610 of the present embodiment. The processing unit (CPU) 622 performs general control operations for data exchange of the memory controller 620.

Although not shown in the drawing, it is obvious to a person skilled in the art to which the embodiment pertains that the memory system 600 in accordance with the embodiment may be additionally provided with a ROM which

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stores code data for interfacing with the host. The nonvolatile memory device (NVM Device) 610 may be provided as a multi-chip package which is constituted by a plurality of flash memory chips.

The memory system 600 in accordance with the embodiment, described above, may be provided as a storage medium of high reliability, which has a low probability of an error to occur. In particular, the nonvolatile memory device of the present embodiment may be included in a memory system such as a solid state disk (SSD) which is being actively studied recently. In this case, the memory controller 620 may be configured to communicate with an exterior (for example, the host) through one of various interface protocols such as a USB (universal serial bus) protocol, an MMC (multimedia card) protocol, a PCI-E (peripheral component interconnection express) protocol, an SATA (serial advanced technology attachment) protocol, a PATA (parallel advanced technology attachment) protocol, an SCSI (small computer system interface) protocol, an ESDI (enhanced small disk interface) protocol and an IDE (Integrated Drive Electronics) protocol.

FIG. 13 is a block diagram schematically illustrating a representation of a computing system including a memory device in accordance with an embodiment of the disclosure.

Referring to FIG. 13, a computing system 700 in accordance with an embodiment may include a memory system 710, a microprocessor (CPU) 720, a RAM 730, a user interface 740 and a modem 750 such as a baseband chipset, which are electrically coupled to a system bus 760. In the case where the computing system 700 in accordance with the embodiment is a mobile device, a battery (not shown) for supplying the operating voltage of the computing system 700 may be additionally provided. Although not shown in the drawing, it is obvious to a person skilled in the art to which the embodiment pertains that the computing system 700 in accordance with the embodiment may be additionally provided with an application chipset, a camera image processor (CIS), a mobile DRAM, and so on. The memory system 710 may configure, for example, an SSD (solid state drive/disk) which uses a nonvolatile memory to store data. Otherwise, the memory system 710 may be provided as a fusion flash memory (for example, an OneNAND flash memory).

Although exemplary embodiments of the disclosure have been described for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure. Therefore, the embodiments disclosed above and in the accompanying drawings should be considered in a descriptive sense only and not for limiting the technological scope. The technological scope of the disclosure is not limited by the embodiments and the accompanying drawings. The spirit and scope of the disclosure should be interpreted by the appended claims and encompass all equivalents falling within the scope of the appended claims.

What is claimed is:

1. A memory device comprising:

- a first semiconductor structure including a peripheral circuit which is defined in a cell region of a substrate, a plurality of pass transistors which are defined in a row decoder region of the substrate, a first bonding layer which includes a plurality of first bonding pads, and a plurality of bottom wiring layers which are disposed between the substrate and the first bonding layer;
- a second semiconductor structure disposed on the first semiconductor structure, and including a second bond-

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ing layer which includes a plurality of second bonding pads bonded to the plurality of first bonding pads, a memory cell array, and at least one top wiring layer which is disposed between the second bonding layer and the memory cell array; and

a plurality of global lines disposed in the row decoder region, and configured to transfer operating voltages to the plurality of pass transistors,

wherein the plurality of bottom wiring layers include bottom wiring layers of a first tier and bottom wiring layers of a second tier which are disposed over the bottom wiring layers of the first tier, and

wherein the plurality of global lines are disposed in at least one of the bottom wiring layers of the first tier.

2. The memory device according to claim 1, wherein the bottom wiring layers of the first tier include a first bottom wiring layer and a second bottom wiring layer over the first bottom wiring layer,

the first bottom wiring layer is a lowermost bottom wiring layer among the plurality of bottom wiring layers, and the plurality of global lines are disposed in the second bottom wiring layer.

3. The memory device according to claim 2, further comprising:

a plurality of local lines disposed in the row decoder region, and configured to transfer the operating voltages, provided through the plurality of pass transistors, to the memory cell array,

wherein the plurality of local lines comprise a plurality of first local lines which are disposed in the second bottom wiring layer, and a plurality of second local lines which are disposed in the at least one top wiring layer.

4. The memory device according to claim 2, further comprising:

a plurality of coupling lines configured to electrically couple elements included in the peripheral circuit, and disposed in the row decoder region of at least one bottom wiring layer among the bottom wiring layers of the second tier.

5. The memory device according to claim 4, wherein the coupling lines are disposed in at least two bottom wiring layers among the bottom wiring layers of the second tier, and

coupling lines disposed in one of the at least two bottom wiring layers extend in a first direction, and coupling lines disposed in the other of the at least two bottom wiring layers extend in a second direction crossing the first direction.

6. The memory device according to claim 4, further comprising:

a plurality of local pads coupled to the plurality of first local lines, and disposed in the row decoder region of one bottom wiring layer among the bottom wiring layers of the second tier,

wherein the plurality of local pads are disposed in a plurality of rows in a second direction crossing a first direction.

7. The memory device according to claim 6, wherein coupling lines disposed in one bottom wiring layer among the bottom wiring layers of the second tier extend in the second direction,

the one bottom wiring layer includes a first zone and a second zone which are divided by a local pad row, and the coupling lines disposed in the one bottom wiring layer include a first group of coupling lines which are disposed in the first zone, and a second group of coupling lines which are disposed in the second zone.

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8. The memory device according to claim 7, wherein the first group of coupling lines have a first width, and the second group of coupling lines have a second width smaller than the first width.

9. The memory device according to claim 7, wherein signals of the first group of coupling lines and signals of the second group of coupling lines have different characteristics.

10. The memory device according to claim 7, wherein the signals of the first group of coupling lines include at least one of power and an analog signal, and the signals of the second group of coupling lines include a digital signal.

11. The memory device according to claim 6, wherein coupling lines disposed in one bottom wiring layer among the bottom wiring layers of the second tier extend in the first direction, and are disposed alternately with the plurality of local pads in the second direction.

12. A memory device comprising:

a first semiconductor structure including a substrate which includes first and second cell regions adjacent to each other in a first direction and a row decoder region between the first cell region and the second cell region, a peripheral circuit which is defined in the first and second cell regions of the substrate, a plurality of pass transistors which are defined in the row decoder region of the substrate, a first bonding layer which includes a plurality of first bonding pads, and a plurality of at least three bottom wiring layers which are disposed between the substrate and the first bonding layer;

a second semiconductor structure disposed on the first semiconductor structure, and including a second bonding layer which includes a plurality of second bonding pads bonded to the plurality of first bonding pads, a memory cell array, and at least one top wiring layer which is disposed between the second bonding layer and the memory cell array; and

a plurality of global lines disposed in the row decoder region, and configured to transfer operating voltages to the plurality of pass transistors,

wherein the plurality of global lines are disposed in at least one of a first bottom wiring layer which is positioned at a lowermost part among the plurality of bottom wiring layers and a second bottom wiring layer immediately over the first bottom wiring layer.

13. The memory device according to claim 12, further comprising:

a plurality of coupling lines disposed in the row decoder region, and configured to couple elements which are included in the peripheral circuit,

wherein the plurality of bottom wiring layers include a third bottom wiring layer over the second bottom wiring layer, a fourth bottom wiring layer over the third bottom wiring layer, and a fifth bottom wiring layer over the fourth bottom wiring layer, and

wherein the coupling lines comprise:

a plurality of first coupling lines disposed in the second bottom wiring layer;

a plurality of second coupling lines disposed in the third bottom wiring layer;

a plurality of third coupling lines disposed in the fourth bottom wiring layer; and

a plurality of fourth coupling lines disposed in the fifth bottom wiring layer.

14. The memory device according to claim 13, wherein the plurality of global lines, the plurality of first coupling lines and the plurality of fourth coupling lines extend in a second direction crossing the first direction, and

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the plurality of second coupling lines and the plurality of third coupling lines extend in the first direction.

15. The memory device according to claim 14, further comprising:

a plurality of local lines disposed in the row decoder region, and configured to transfer the operating voltages, provided through the plurality of pass transistors, to the memory cell array,

wherein the plurality of local lines comprise:

a plurality of first local lines disposed in the second bottom wiring layer; and

a plurality of second local lines disposed in the at least one top wiring layer.

16. The memory device according to claim 15, further comprising:

a plurality of first local pads disposed in the row decoder region of the third bottom wiring layer, and coupled to the plurality of first local lines through a plurality of first contacts;

a plurality of second local pads disposed in the row decoder region of the fourth bottom wiring layer, and coupled to the plurality of first local pads through a plurality of second contacts; and

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a plurality of third local pads disposed in the row decoder region of the fifth bottom wiring layer, and coupled to the plurality of second local pads through a plurality of third contacts,

wherein the plurality of third local pads are disposed in a plurality of rows in the second direction, and third local pads included in one row are disposed alternately with the plurality of fourth coupling lines in the first direction.

17. The memory device according to claim 16, wherein two third local pads among the plurality of third local pads are coupled in common to one second local pad, and each of the plurality of third local pads is coupled to one first bonding pad.

18. The memory device according to claim 17, wherein the two third local pads which are coupled in common to the one second local pad are adjacent to each other with one fourth coupling line interposed therebetween.

19. The memory device according to claim 17, wherein two second bonding pads are coupled in common to each of the plurality of second local lines.

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